



Contents lists available at ScienceDirect

Tunnelling and Underground Space Technology incorporating Trenchless Technology Research

journal homepage: www.elsevier.com/locate/tust

Seg2Tunnel: A hierarchical point cloud dataset and benchmarks for segmentation of segmental tunnel linings

Wei Lin^{a,b,c,d}, Brian Sheil^b, Pin Zhang^b, Biao Zhou^{a,c,d}, Cheng Wang^e, Xiong Yao Xie^{a,c,d,*}^a Department of Geotechnical Engineering, College of Civil Engineering, Tongji University, Shanghai 200092, China^b Laing O'Rourke Centre for Construction Engineering and Technology, Department of Engineering, University of Cambridge, Cambridge CB3 0FA, United Kingdom^c State Key Laboratory of Disaster Reduction in Civil Engineering, Tongji University, Shanghai 200092, China^d Key Laboratory of Geotechnical and Underground Engineering of the Ministry of Education, Tongji University, Shanghai 200092, China^e State Grid Shanghai Municipal Electric Power Company, Shanghai 200122, China

ARTICLE INFO

Keywords:

Segmental tunnel lining
Deep learning
Point cloud
Segmentation
Computer vision

ABSTRACT

Point clouds provide a novel and effective alternative to understanding the structural behaviours of segmental tunnel linings. 3D deep learning (DL) has emerged as a promising technology capable of automatically deriving point-wise semantic and instance labels from point clouds. The utilisation of 3D DL in segment segmentation of tunnel point clouds has not been explored and the development of tailored 3D DL networks has been hindered by the absence of specialised datasets and benchmarks. To bridge this gap, this paper introduces a richly annotated hierarchical dataset: 'Seg2Tunnel', acquired from five tunnels and including 1,300 tunnel rings. Using the Seg2Tunnel dataset, the feasibility of applying 3D DL to the segment segmentation is demonstrated for the first time. Experiments are conducted to investigate the influences of training set size, data augmentation strategy, input size, and hyperparameter on the performance of trained 3D DL models and to provide benchmarks and insights for future uses of the Seg2Tunnel dataset. The 3D DL models trained by the Seg2Tunnel dataset outperform currently existing image- and voxel-based DL methods. The Seg2Tunnel dataset and benchmarks are fundamental in shaping the design of 3D DL networks tailored for tunnel point clouds. The study provides a novel paradigm for automatically understanding the tunnel structural elements in the point clouds, paving the way for unmanned construction and intelligent evaluation of segmental tunnel linings.

1. Introduction

The growing need to exploit underground space for urban infrastructure development has prompted substantial increases in shield tunnel construction worldwide. Construction quality is a key performance indicator for mechanised tunnelling (Chang et al., 2023; Huang et al., 2021), especially in light of the rising promise of unmanned construction. On the other hand, a substantial amount of underground infrastructure is also nearing the end of its life, such that structural health becomes a critical concern to ensure the day-to-day functioning of cities, including transportation and energy transmission (Zhou et al., 2014).

Segmental linings are one of the most common permanent tunnel support structures. Complex and stochastic soil-structure interactions and frequent adjacent construction have been shown to cause unpredictable and randomly distributed deformations of segmental linings

(Chen et al., 2020; Lin et al., 2023a, 2022b; Yu et al., 2023). Such deformations can also be induced by mechanical construction errors (Xu et al., 2019; Zhou et al., 2018, 2017). Hence, it is important to inspect and evaluate the segments during both construction and service.

Segmental linings are also problematic for asset management due to the large number of segments to be inspected, which can reach thousands per kilometre (Hegemann et al., 2020). Prerequisites for high-fidelity structural analysis of these linings include (a) a detailed description of the segment and joint arrangements and (b) the current (deformed) geometry (Xie et al., 2023). For the latter, advanced measurement technologies, such as wireless tilt sensors (Chang et al., 2022) and optic fibre sensors (Mu et al., 2021), have been deployed to measure the deformation of segments and joints. However, such point-wise measurements do not fully describe the complex deformation field (Wang et al., 2021). Hence, comprehensive end-to-end inspection techniques for large-scale tunnels are required.

* Corresponding author at: Department of Geotechnical Engineering, College of Civil Engineering, Tongji University, Shanghai 200092, China.

E-mail address: xiexiongyao@tongji.edu.cn (X. Xie).

<https://doi.org/10.1016/j.tust.2024.105735>

Received 4 November 2023; Received in revised form 14 February 2024; Accepted 27 March 2024

Available online 4 April 2024

0886-7798/© 2024 Elsevier Ltd. All rights reserved.



Fig. 1. Field data acquisition of the Seg2Tunnel dataset.

Laser scanning is a powerful technology to capture rich geometric information about the surrounding environment (Xie and Lu, 2017). Point clouds acquired by laser scanning are valuable digital assets for shield tunnels, which can be used for 3D deformation measurement (Du et al., 2022; Lin et al., 2023b, 2022a; Yi et al., 2019; Yu et al., 2021) and defect detection (Huang et al., 2018; Zhou et al., 2021). A prerequisite for automatic extraction of present-day deformation measurements is the identification of individual segments from the background scene (Cui et al., 2019). The lack of semantic information in point clouds means that such data cannot be processed automatically and thus structural analysis workflows are still highly manual and time-consuming (Liebers et al., 2023). Heuristic algorithms using prior knowledge or experience of crafted point cloud features have been developed to facilitate point cloud segmentation of segmental linings (Li et al., 2022; Xie and Lu, 2017). However these heuristic algorithms suffer from several drawbacks: (a) their performance is severely affected by missing point cloud features, (b) their reliability is sensitive to the scanning environment, (c) their optimisation is cumbersome since heuristic algorithms consist of multiple components (Duan et al., 2021; Li et al., 2022), and (d) original design documents are often needed for certain algorithms (Yi et al., 2019).

Hence, an accurate and robust methodology for automatic analysis of point clouds is essential for improving (a) lining construction quality control and (b) lining serviceability evaluation. The emergence of deep learning (DL) technology has already shown considerable promise for improving the efficiency and accuracy of tunnel management (Xie et al., 2022; Zhang et al., 2022b). 3D DL networks, taking 3D point clouds as

the direct inputs (Qi et al., 2017), have shown an improved ability to generalise compared to heuristic algorithms owing to improved feature design and selection abilities (Soilán et al., 2020). Although recent research has explored the application of 3D DL networks to tunnel point cloud segmentation tasks (Ji et al., 2023, 2022; Zhou et al., 2023), segment segmentation using 3D DL has not been achieved. A key requirement is access to training datasets, as the performance of DL models depends heavily on the amount and quality of such data (Schroff et al., 2015). The roles of annotated datasets in both model training and evaluation are significant. Several renowned generic datasets, such as ShapeNet (Chang et al., 2015), PartNet (Mo et al., 2019), S3DIS (Armeni et al., 2017), Semantic3D (Hackel et al., 2018), and SemanticKITTI (Behley et al., 2021), have fostered the development of many renowned 3D DL networks (Xie et al., 2020). In contrast, there is a distinct lack of annotated datasets for tunnel point clouds which presents a significant obstacle for 3D DL segmentation. Thus, specialised datasets and their corresponding benchmarks are urgently required for the development of 3D DL for tunnel point clouds.

To that end, this paper introduces a new and richly annotated segmental tunnel lining point cloud dataset, 'Seg2Tunnel', to enable automated segment segmentation using 3D DL for the first time. The Seg2Tunnel dataset is designed with a hierarchical architecture to allow rigorous training and evaluation of segmentation performance. This study also provides proof of concept and corresponding benchmarks by applying state-of-the-art 3D DL networks to the Seg2Tunnel dataset. Data experiments are presented to explore the influences of key influencing factors on the 3D DL model performance, namely, training set size, data augmentation strategy, input size, and hyperparameter. To realise the fully automatic understanding of large-scale tunnel point clouds, a novel two-stage framework integrating both semantic and instance segmentation networks is ultimately proposed. Deformation calculation results based on automatically segmented point clouds are provided to further validate the effectiveness of the proposed methodology.

2. Dataset methodology

2.1. Data acquisition

The point clouds in the Seg2Tunnel dataset are acquired from both as-built (before service) and in-service shield tunnels from Wuxi, Shanghai, and Fuzhou, using terrestrial laser scanners (TLS), as shown in Fig. 1. The TLS used for scanning includes the Leica C10 and Leica P40; both have a scanning noise (root mean square error) of <0.5 mm for

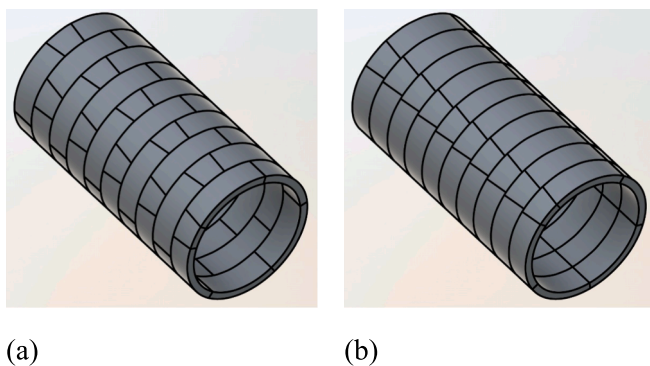


Fig. 2. Schematic illustration of the two assembly types: (a) staggered joints and (b) continuous joints.

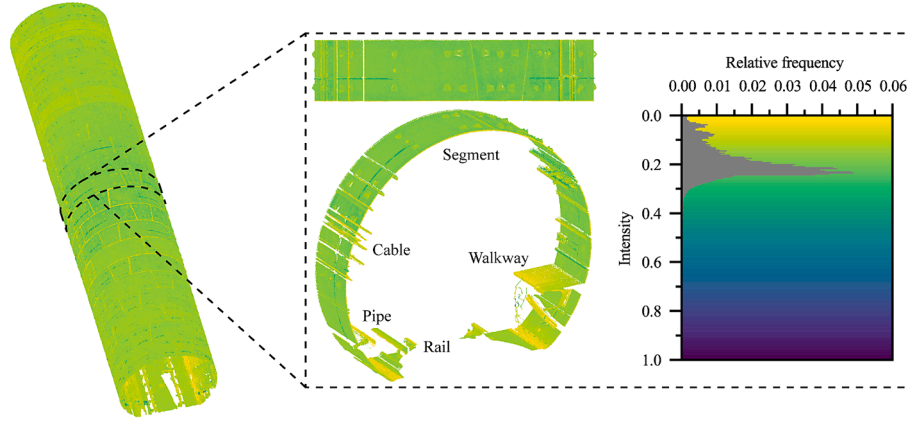
Table 1

Details of the scanned shield tunnels in the Seg2Tunnel dataset.

Tunnel No.	Scanned ring count	Scanned length/m	Inner diameter/m	Ring length/m	Number of segments per ring	Number of scanning stations	Tunnel location	Assembly type*	Tunnel status	Laser scanner
T1	434	520.8	5.5	1.2	6	17	Wuxi	Staggered	In-service	Leica C10
T2	304	364.8	5.5	1.2	6	14	Wuxi	Staggered	In-service	Leica C10
T3	119	142.8	5.5	1.2	6	4**	Shanghai	Continuous	In-service	Leica P40
T4	342	615.6	7.5	1.8	7	17	Fuzhou	Staggered	As-built	Leica C10
T5	101	181.8	7.5	1.8	7	7	Fuzhou	Staggered	As-built	Leica C10

*See Fig. 2.

**For T3, the four individual point clouds were assembled into one.

**Fig. 3.** An exemplar scanned point cloud from tunnel T1 in the Seg2Tunnel dataset.

scanning distances < 50 m. In addition to the tunnel structural elements, each point cloud also includes cables, pipes, rails, walkways, and other non-structural elements. Hence, all possible elements and noise that may affect the point cloud segmentation are considered in the present dataset.

Shield tunnels are hierarchically composed of segments and rings. For metro shield tunnels, each ring is usually assembled into six or seven segments, including one key segment, two adjacent segments, and three or four standard segments. The segments and rings are connected by longitudinal joints and circumferential joints respectively. There are two common types of segment assembly: staggered joints and continuous joints (Liu et al., 2017), as shown in Fig. 2. The Seg2Tunnel dataset comprises five shield tunnels, denoted 'T1' to 'T5', and contains both segment assembly types, different tunnel diameters, and different numbers of segments (see Table 1). For each tunnel, the point clouds are acquired from different scanning stations without overlaps. For tunnel T3, the point clouds were manually assembled into one point cloud using registration targets between scanning stations (implemented in the software 'Cyclone'). The total number of scanned rings is 1,300 equating to a total length of 1,825.8 m. The total number of scanned points is 821,307,768. An exemplar point cloud is displayed in Fig. 3. The acquired points have four features: the three Cartesian coordinates and the intensity value (optical power per unit area). The colour in Fig. 3 denotes the intensity of each point, ranging from 0 to 1, of which the distribution is also provided for the present ring.

2.2. Data hierarchization

The structure of shield tunnels lends itself to being characterized by a hierarchical assembly comprising three levels: segment level, ring level, and 'scene' level (i.e. that captured by one scanning station) (Ninic et al., 2021). The 3D DL tasks for segmental lining point clouds can thus be divided into two levels, namely semantic segmentation and instance segmentation. The 3D DL networks also have different features, some are designed for object-level point clouds (Qi et al., 2017) and some are

designed for scene-level point clouds (Hu et al., 2022). Hence, to enable rigorous future model training and evaluation on Seg2Tunnel, the dataset is designed with multilevel data hierarchization.

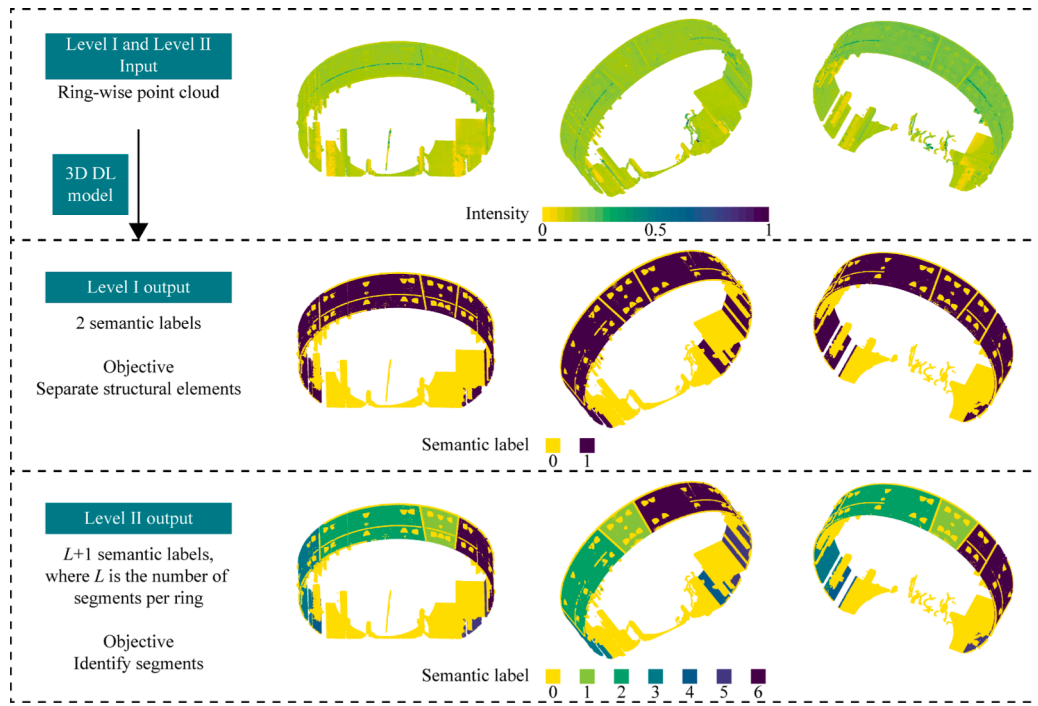
The four-level data hierarchization of the Seg2Tunnel dataset is shown in Fig. 4. The different labels are presented in different colours in Fig. 4. Label 0 denotes non-structural elements and noise in all cases. The points corresponding to label 0 are the same for all four levels. For Level I, the structural elements, i.e., segments, are lumped together and annotated as label 1. For all other levels, the different segments per ring are individually assigned non-zero labels (from 1 to $L + 1$ where L is the number of segments per ring). For Level IV, the structural elements are also assigned instance labels.

For Level I and Level II segmentation, the input is the ring-wise point cloud. Level I segmentation seeks to separate the structural elements of interest (i.e. the segmental linings) from all other elements and noise. Level II not only segments the structural elements from the background but also annotates each segment individually such that the models can be trained to recognise different segments, ultimately enabling future ambitions of automatically calculating the deformation of each individual segment and joint.

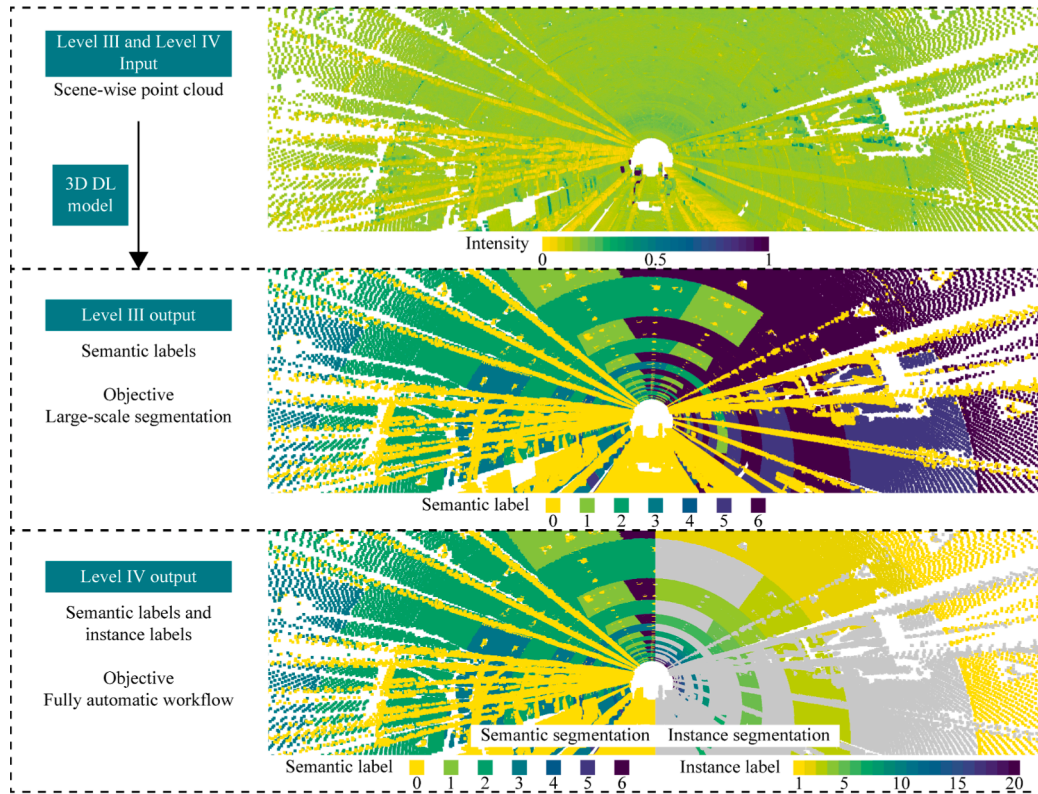
Rather than segmenting a tunnel ring by ring, a more efficient and practical implementation is to supply the networks with a full scene comprising an unspecified number of rings (Level III and Level IV). Thus, Level III seeks to segment a scene-wise point cloud semantically. Level IV is the most advanced segmentation task, seeking to identify both semantics and instances from a scene automatically. This bottom-up data hierarchization with increasing difficulty of implementation, from coarse segmentation to fine segmentation, is pursued to allow assessment of the maturity of 3D DL networks in tunnel point clouds.

2.3. Data annotation

To achieve rigorous model training and evaluation, the data annotation needs to be comprehensive, consistent, and accurate for the specified labels (Everingham et al., 2010). Instead of pixel-wise or voxel-



(a)



(b)

Fig. 4. Illustration of the hierarchical data architecture adopted for the Seg2Tunnel dataset: (a) ring-wise: Level I and Level II and (b) scene-wise: Level III and Level IV.

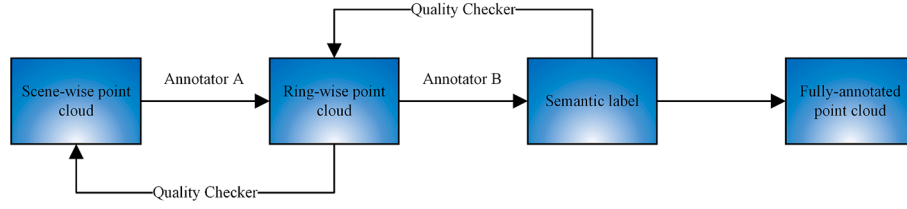


Fig. 5. Illustration of the adopted data annotation workflow.

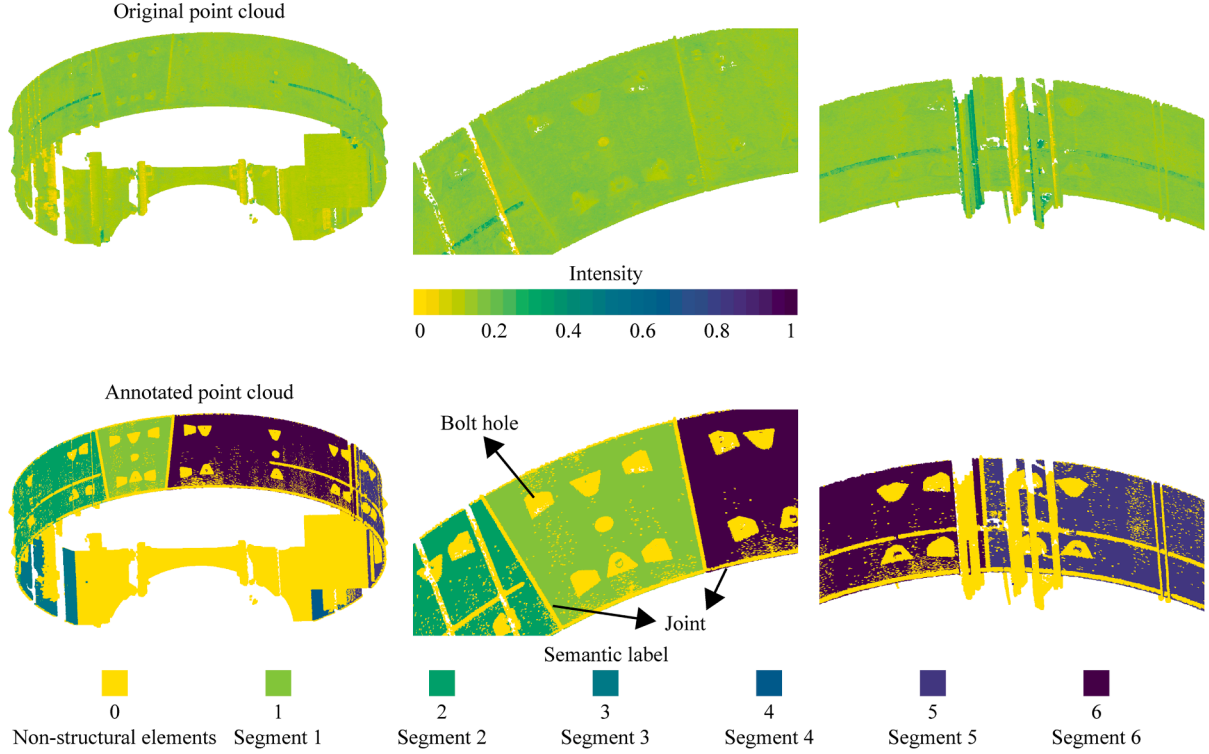


Fig. 6. Exemplar Level II annotated point cloud (T2-4-301) from the Seg2Tunnel dataset: (a) original point cloud and (b) annotated point cloud.

wise annotation, every single point in Seg2Tunnel is labelled individually. This exhaustive manual annotation of the ground truth dataset avoids any biases introduced by particular segmentation algorithms (Hackel et al., 2018). The annotators are trained to understand the structure of the tunnel segmental linings using design and construction documents. Annotation is performed by considering both geometric and intensity features as shown in Fig. 3.

In keeping with the adopted hierarchical architecture, annotation follows a top-down workflow, as shown in Fig. 5. First, the scene-wise point clouds are segmented into ring-wise point clouds. Then, tunnel lining segments are separated from all other elements and noise for each separate ring point cloud. Points relating to joints and bolt holes require careful treatment and are segmented from the point clouds. One group of annotators is tasked with ring segmentation and another group is

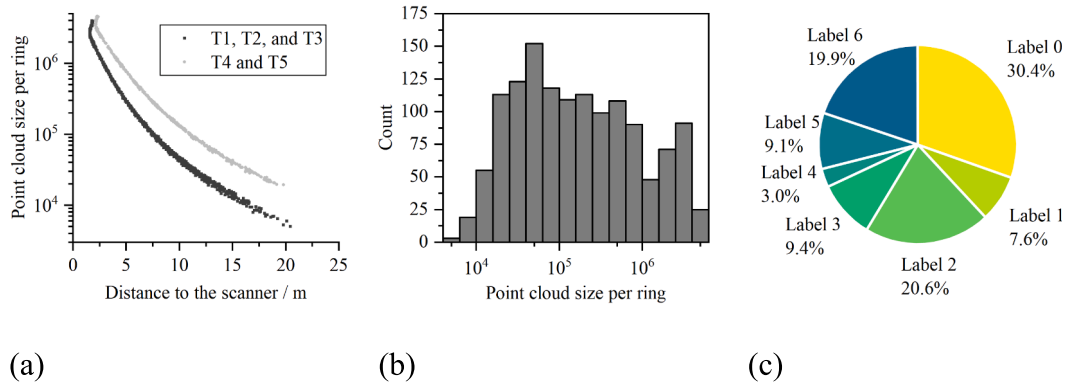


Fig. 7. Point cloud size distribution of the Seg2Tunnel dataset: (a) point cloud size per ring as a function of the distance to the scanner, (b) histogram of the point cloud size per ring, and (c) class proportion for all point clouds for Level II to Level IV.

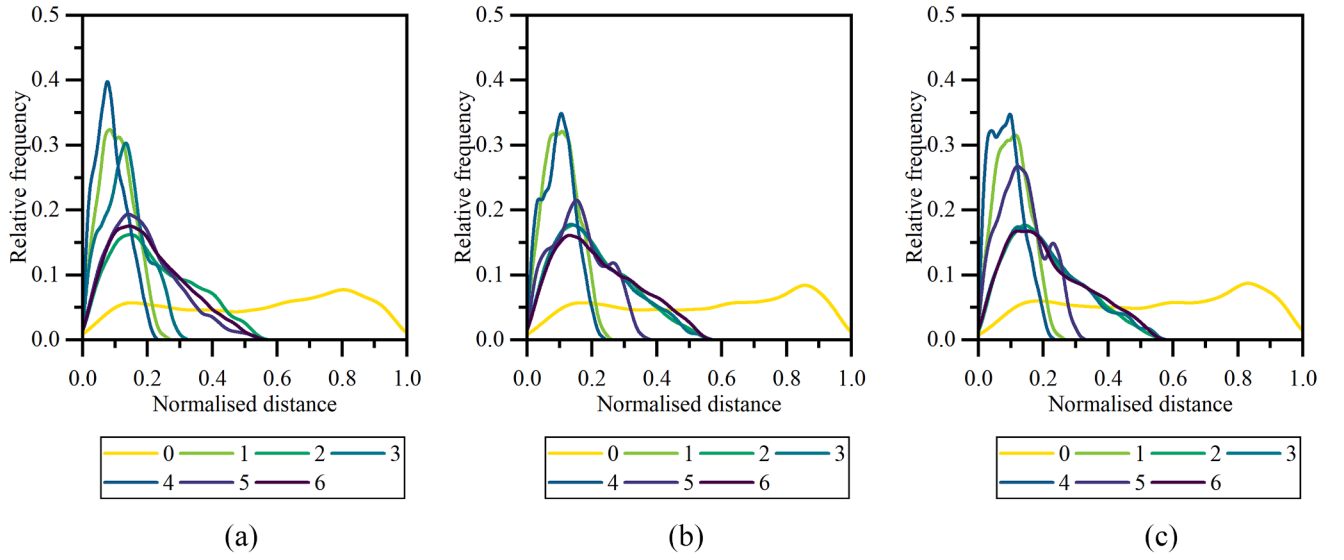


Fig. 8. The shape signature of each class in the point cloud per ring for Level II to Level IV for three randomly selected rings: (a) T1-8-467, (b) T1-16-966, and (c) T2-9-694.

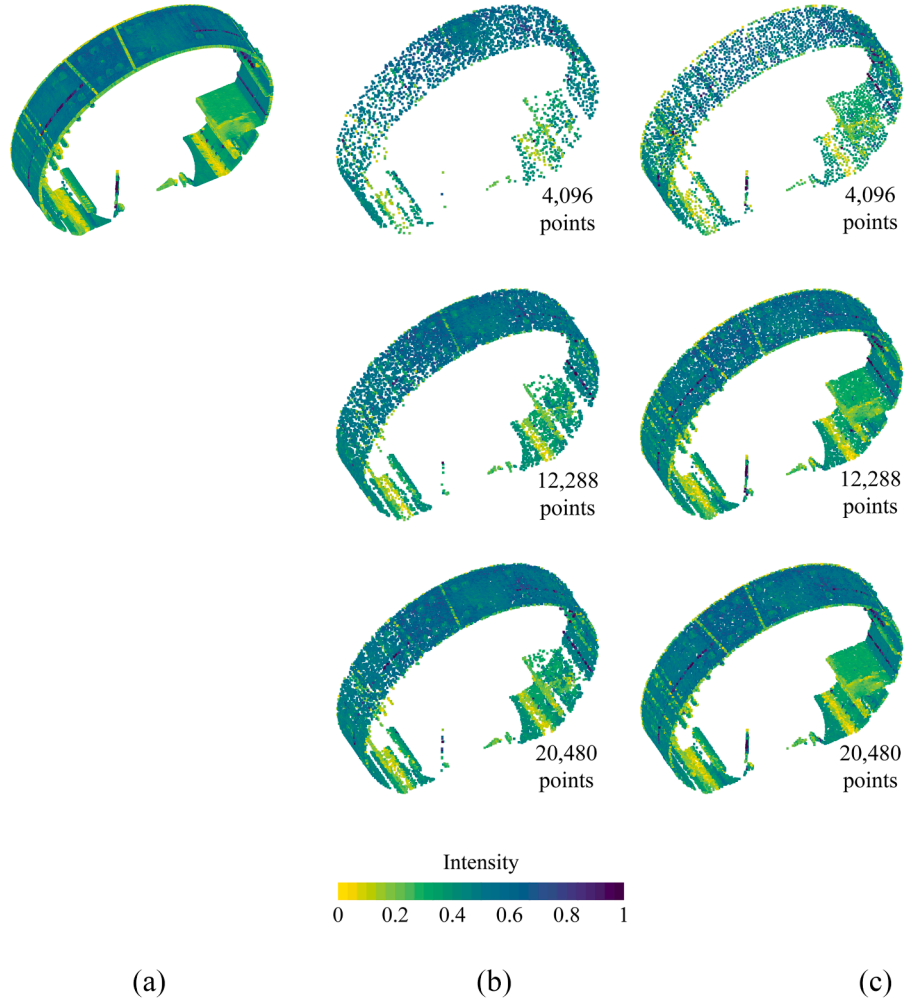


Fig. 9. Sampling results of the ring-wise point cloud from the Seg2Tunnel dataset (ring T1-1-20 chosen for illustrative purposes): (a) raw point cloud, (b) RS, and (c) FPS.

tasked with segment segmentation. The accuracy of every point cloud is then checked and confirmed by the authors for any residual annotation errors. Fig. 6 presents an exemplar Level II annotated point cloud, where different colours denote different labels. The segments are labelled sequentially according to their layout in the lining ring, starting from the key segment. It is worth mentioning that there may be minor annotation noises due to inevitable annotation mistakes, the presence of which can be seen in Fig. 6 (scattered points with label 0), though they are uncommon in the Seg2Tunnel dataset. All 1,300 rings are annotated to Level I. Finer segmentation (Level II to Level IV) is performed on 738 rings from tunnels T1 and T2. The annotated data are all organized from Level I to Level IV. The file naming structure ‘Tl-m-n’ is adopted where l , m and n refer to the tunnel number, scanning station number, and ring number, respectively.

3. Dataset statistics

The resulting Seg2Tunnel dataset is analysed to provide quantitative insights into its key characteristics. The point clouds in Seg2Tunnel are acquired by TLS so the point density decreases with distance to the scanner, as shown in Fig. 7(a). Tunnels with similar diameters have similar point density distributions e.g. the diameter of tunnels T1, T2, and T3 is 5.5 m whereas that of tunnels T4 and T5 is 7.5 m. The histogram in Fig. 7(b) demonstrates that the individual ring-wise point clouds from the Seg2Tunnel dataset vary significantly in size from thousands of points to millions. For Level II to Level IV, the point cloud proportion of each class is presented in Fig. 7(c), showing that the Seg2Tunnel dataset is ‘class-imbalanced’ where label 0 accounts for the greatest proportion of the point clouds. The number of points with label 4 is less than others since those segments are usually installed at the bottom of the tunnel and are thus often not observable by the scanner due to the intervening pavement.

Fig. 8 plots the ‘shape signature’ (Mahmoudi and Sapiro, 2009; Osada et al., 2001) for each class, calculated as the frequency distribution of the distance between two randomly sampled points with the same label. As a result, the shape signature, represented by the frequency distribution, quantifies the shape of a given point set. The results are normalised to lie in the interval 0 to 1 and are presented for three randomly selected rings. From Fig. 8(a), it can be observed that there is good shape similarity between the segment classes but notable differences in the scale, except for the non-structural element and noise class (label 0). Comparing the results for different rings in Fig. 8(a) to (c), shape similarity is also observed. The result shows that the segments are highly similar for those either in the same ring or in different rings. This similarity highlights the difficulty in identifying individual segments using 3D DL.

4. Benchmarking methodology

4.1. Data sampling and augmentation

Although point intensity theoretically can take a value between 0 and 1, in reality, the intensities are distributed in a narrow interval (see Fig. 3) which limits its usefulness as a feature for segmentation. Thus, point intensities are augmented by applying a min–max scaler to normalise all values as follows:

$$i_a = \begin{cases} 0, & i_0 < i_{\min} \\ \frac{i_0 - i_{\min}}{i_{\max} - i_{\min}}, & i_{\min} \leq i_0 \leq i_{\max} \\ 1, & i_0 > i_{\max} \end{cases} \quad (1)$$

where i_0 is the original intensity value, i_a is the augmented original value, and $i_{\min}$ and $i_{\max}$ are prescribed thresholds (the 1 % percentile and 99 % percentile respectively are adopted here). For Level I and Level II, the Cartesian coordinates of the points are all normalised into the

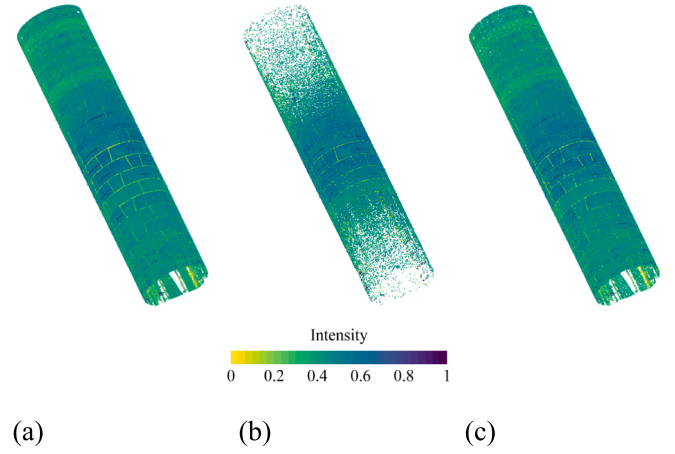


Fig. 10. Sampling results of the scene-wise point cloud from the Seg2Tunnel dataset (station T1-1 chosen for illustrative purposes): (a) raw point cloud, (b) RS, and (c) FPS.

interval -1 to 1 to eliminate scale effects.

It is both common and necessary to sample the point clouds before DL training and evaluation to improve computational efficiency, whilst also minimizing the effect on accuracy. Sampling is also widely adopted within the 3D DL networks, usually serving as components of the encoders. Hence, two popular sampling strategies — ‘random sampling’ (RS) and ‘farthest point sampling’ (FPS) (Qi et al., 2017), — are compared using the Seg2Tunnel dataset. As the name suggests, RS involves randomly selecting individual points for removal where each point has the same probability of being sampled. FPS iteratively returns the farthest point from the sampled points and performs distance updating.

The results of applying RS and FPS to the raw point clouds are shown in Fig. 9 and Fig. 10 for a given tunnel ring and tunnel scene respectively. For the ring-wise point clouds, it can be seen that FPS returns a more even point distribution and thus preserves more point cloud details such as joints and bolt holes (see Fig. 9). The influence of sampling size is also considered, where 12,288 points sampled by FPS — compared to 3,694,229 points in the raw point cloud — appear to provide a reasonable balance between size and fidelity. Similar observations can be deduced for the scene-wise point clouds in Fig. 10. The proportion of point clouds after sampling is plotted in Fig. 11. The results show that the raw data acquisition by TLS suffers from uneven point density, which is then preserved by RS. The computational complexity of FPS and RS algorithms is $O(n^2 n_s)$ and $O(n_s)$ respectively, where n and n_s denote the number of original points and sampled points respectively (Hu et al., 2022). A larger n will significantly increase the computational complexity of FPS, making it prohibitively expensive. By way of example, the computational time for FPS to sample 100,000 points from 1,000,000 points is $\sim 50,000$ times greater than that of RS.

Data augmentation is one of the most important strategies to boost the performance of 3D DL networks (Qian et al., 2022). For 3D data, rotation is the most common augmentation method (Chen et al., 2019). The structure of the tunnel is such that augmentation by ring-wise splitting is also possible. The raw point cloud is repeatedly sampled (RS is adopted here considering the computational efficiency) without replacement by splitting into several slightly different point clouds. A parameter, denoted as the splitting count factor, is used to control the number of split point clouds. Then random rotation is applied to all points of each split point cloud:

$$\begin{pmatrix} x_r \\ y_r \\ z_r \end{pmatrix} = R_x(\alpha) R_y(\beta) R_z(\gamma) \begin{pmatrix} x_0 \\ y_0 \\ z_0 \end{pmatrix} \quad (2)$$

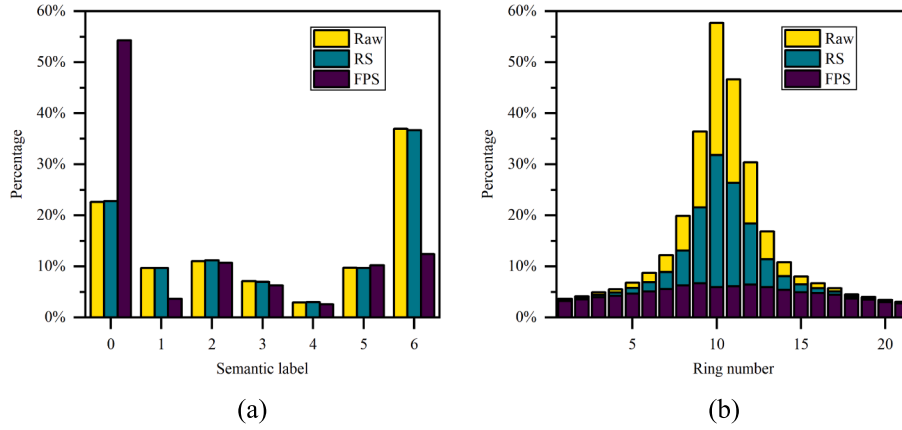


Fig. 11. Point cloud proportion achieved using different sampling strategies: (a) ring-wise point cloud (T1-1-20) and (b) scene-wise point cloud (T1-1).

Table 2

Learning rate and scheduler of model training.

DL network	Learning rate	Scheduler ("PyTorch documentation," 2023)	Number of epochs
PointNet++	0.001	StepLR (step_size = 20 and gamma = 0.7)	250
RandLA-Net	0.01	ExponentialLR (gamma = 0.95)	100
Mask3D	0.0001	OneCycleLR	1000
TD3D	0.001	MultiStepLR (milestones = [280, 320] and gamma = 0.1)	33

Table 3

The test set of the Seg2Tunnel dataset.

Tunnel No.	Test set	Test set proportion
T1	T1-4, T1-12, and T1-15	3 out of 17 stations
T2	T2-2, T2-10, and T2-14	3 out of 14 stations
T3	T3-1-96 to T3-1-119	23 out of 119 rings
T4	T4-3, T4-6, and T4-17	3 out of 17 stations
T5	T5-1	1 out of 7 stations

2021; Jing et al., 2022). For the instance segmentation task of Level IV, Mask3D, TD3D, and three unsupervised clustering algorithms — DBSCAN (Zhang et al., 2022a), Affinity Propagation (Goyal et al., 2012), and K-means (Itakura et al., 2022) — are compared.

The adaptive moment estimation (Adam) optimiser is adopted for all aforementioned 3D DL networks. The adopted learning rates and schedulers follow recommendations from previous literature (Hu et al., 2022; Kolodiazhnyi et al., 2023; Qi et al., 2017; Schult et al., 2023) and are listed in Table 2. For semantic segmentation, the commonly used cross-entropy loss function is adopted. For instance segmentation, the adopted loss functions follow the implementations in the corresponding literature (Kolodiazhnyi et al., 2023; Schult et al., 2023).

All models were trained using NVIDIA A100-SXM-80 GB GPUs. Data shuffling is employed to ensure the credibility of model training and evaluation (Qi et al., 2017). The training set and test set are split randomly 80:20 (respectively), as shown in Table 3. As the point clouds from tunnel T3 are merged into one point cloud, the last 23 rings are selected as the test set. All the evaluation metrics of trained models are calculated on the segmentation results of the test set.

For semantic segmentation, OA (overall accuracy; Equation (6)), IoU (interaction over union; Equation (7)) and mIoU (mean interaction over union; Equation (8)) are employed to evaluate model performance. In particular, mIoU presents the most reasonable evaluation for class-imbalanced datasets (see Fig. 6) and is thus adopted as the priority metric.

$$OA = \frac{\sum_{j=1}^M (TP)_j}{N} \quad (6)$$

$$IoU = \frac{TP}{TP + FP + FN} \quad (7)$$

$$mIoU = \frac{1}{M} \sum_{j=1}^M IoU_j \quad (8)$$

where M is the number of classes, N is the number of points, and TP , FP , and FN denote true positive, false positive, and false negative predictions respectively.

$$R_z(\alpha) = \begin{pmatrix} \cos\alpha & -\sin\alpha & 0 \\ \sin\alpha & \cos\alpha & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad (3)$$

$$R_y(\beta) = \begin{pmatrix} \cos\beta & 0 & \sin\beta \\ 0 & 1 & 0 \\ -\sin\beta & 0 & \cos\beta \end{pmatrix} \quad (4)$$

$$R_x(\gamma) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos\gamma & -\sin\gamma \\ 0 & \sin\gamma & \cos\gamma \end{pmatrix} \quad (5)$$

where x_0, y_0 , and z_0 denote the original Cartesian coordinates, x_r, y_r , and z_r denote the rotated Cartesian coordinates, α, β , and γ are random rotation angles along axes z, y , and x respectively, and $R_z(\alpha), R_y(\beta)$, and $R_x(\gamma)$ are the rotation matrix along axes z, y , and x respectively.

4.2. Model training and evaluation

'PointNet' and its extension 'PointNet++' (Qi et al., 2017) whereby 3D DL models are trained directly on point clouds. To improve computational efficiency, 3D DL networks tailored to large-scale point clouds have also been proposed, such as the popular 'RandLA-Net' (Hu et al., 2022) and its derivations (Cheng et al., 2023; Zhan et al., 2023). RandLA-Net adopts RS in its encoder while PointNet++ adopts FPS, making the former more suitable for large-scale point clouds. For instance segmentation, there are two types of 3D DL networks: (a) bottom-up methods and (b) top-down methods, represented by 'Mask3D' (Schult et al., 2023) and 'TD3D' (Kolodiazhnyi et al., 2023) respectively.

3D DL networks with appropriate architectures are selected to provide suitable benchmarks for the Seg2Tunnel dataset. The benchmark adopted for Level I and Level II is PointNet++. For the large-scale scene-wise point clouds of Level III, RandLA-Net is selected as the benchmark for its capability to handle large-scale point clouds (Deng and Dong,

Table 4

Model training configurations adopted for the data experiments.

Level	Investigated factor	Other factors			
		Data augmentation strategy	Training set size	Input size	Hyperparameter
Level I	Training set size	(a) 2D rotation(b) Splitting count factor 6	/	8,192	Default*
Level II	Data augmentation strategy	/	Full training set	8,192	Default*
	Training set size	(a) 2D rotation(b) Splitting count factor 12	/	12,288	Default*
	Input size	(a) 2D rotation(b) Splitting count factor 12	Full training set	/	Default*
	Hyperparameter	(a) 2D rotation(b) Splitting count factor 12	Full training set	8,192	/
Level III	Input size	(a) 2D rotation(b) Splitting count factor 3	Full training set	/	Default**
	Hyperparameter	(a) 2D rotation(b) Splitting count factor 3	Full training set	40,960	/

*The hyperparameters are set the same as the source code of PointNet++ (Qi et al., 2017).

**The hyperparameters are set the same as the source code of RandLA-Net (Hu et al., 2022).

To evaluate the instance segmentation performance, the widely used mAP (mean average precision) is employed:

$$mAP = \frac{1}{M} \sum_{j=1}^M AP_j \quad (9)$$

where M is the number of classes and AP (average precision) is calculated by the method proposed in the COCO dataset (Lin et al., 2014; Yan et al., 2023). mAP is determined using an IoU threshold varying from 50 % to 95 % with a step of 5 %. Another two metrics, mAP@50 and mAP@25, are also considered and represent the scores obtained with IoU thresholds of 50 % and 25 %, respectively.

4.3. Data experiment

Different influencing factors for model performance are focused on each level, including the training set size, data augmentation strategy, input size, and hyperparameter. In Level I experiments, the training set size (number of point clouds) is varied from one tunnel (T1) to five tunnels (T1 to T5) to explore the importance of training set diversity. Similar experiments are considered on Level II, with the training set size increasing from 1 to 25 scanning stations. Both 2D rotation and 3D rotation are considered for data augmentation, using Equation (2)–Equation (5). For 2D rotation, α is randomly generated while β and γ are set as 0 to rotate the point clouds randomly on the horizontal (x-y) plane only. For 3D rotation, α , β , and γ are all randomly generated. The splitting count factors considered include 1, 3, 6, 12, and 18. Larger input sizes (number of points in each point cloud) may correspond to better model performance but with higher annotation and computational cost. For ring-wise point clouds, the compared input size varies from 2,048 to 24,576 points. For large-scale point clouds on Level III, five input sizes, from 30,720 to 71,680, are compared. Hyperparameters of 3D DL networks are also important for point cloud segmentation. For PointNet++, there are three main types of hyperparameters to be investigated, namely neighbour radius, neighbour number, and multi-layer perceptron (MLP) size (Qi et al., 2017). Similarly, for RandLA-Net, the main hyperparameter of interest is the neighbour numbers (Hu et al., 2022) where values of 4, 8, 16, 24, and 28 are considered. The experiment configurations of Level I to Level III are listed in Table 4.

3D instance segmentation (for Level IV), remains a challenging task in computer vision. The AP@50 of the state-of-the-art network on the S3DIS dataset is 0.755 (Schult et al., 2023). The available choices of 3D DL networks for 3D instance segmentation are hence limited. The first thing to consider in Seg2Tunnel Level IV experiments is the feasibility of 3D DL networks. Two 3D instance segmentation tasks are designed to investigate feasibility: (a) Task 1: instance segmentation of whole point clouds and (b) Task 2: instance segmentation of partial point clouds with individual labels. For Task 1, the instances of all the classes in the point clouds are segmented by one trained model in one go, which means that the 3D DL model is trained for not only semantic segmentation but also instance segmentation. For Task 2, the partial point clouds with individual semantic labels are separately input into the 3D DL models for

Table 5

Model performance of PointNet++ on Level I with different training set sizes.

Model No.	Training set	Training set size	OA	mIoU
V1	T1	1,670	0.890	0.789
V2	T1 and T2	2,906	0.907	0.811
V3	T1, T2, and T3	3,476	0.933	0.854
V4	T1, T2, T3, and T4	5,017	0.960	0.906
V5	T1, T2, T3, T4, and T5	5,519	0.960	0.907

The bold font denotes the highest metric in a set of data experiment.

instance segmentation, provided that the semantic labels have been predicted. Two state-of-the-art networks — Mask3D and TD3D — are adopted for the two tasks. The three unsupervised clustering algorithms are also applied to Task 2 for additional comparison. The parameters of the unsupervised clustering algorithm are all adjusted for the best segmentation results. It is worth noting that the models are tested on the same test set when conducting the data experiments to achieve rigorous comparison.

Since the end goal of Seg2Tunnel is to facilitate structural analysis of segmental linings, the error of the deformation calculation on Level I is also provided as an expanded evaluation of the segmentation performance. The convergence deformations of each ring can be calculated by sequentially performing axis fitting, projection, and elliptical fitting (Xie and Lu, 2017). As a result, the fitted ellipse, determined by the least square method, is used to calculate the maximum and minimum convergence, denoted $d_{\max}$ and $d_{\min}$ respectively, by

$$d_{\max} = a - r \quad (10)$$

$$d_{\min} = b - r \quad (11)$$

where a and b are the major and minor semi-axis lengths of the fitted ellipse respectively and r is the design inner radius of the segmental lining rings. The deformation calculation from the manually annotated point clouds is considered the ground truth, owing to its well-documented reliability (Lin et al., 2023b, 2022a; Roca-Pardiñas et al., 2014; Yi et al., 2019).

5. Benchmarking results

5.1. Level I experiment

By way of benchmarking, and to assess the generalisability of 3D DL models trained on Seg2Tunnel, PointNet++ is trained using different training set sizes to produce five different models, as presented in Table 5. As more point clouds are added to the training dataset, there is a notable increase in both OA and, in particular, mIoU. When the point clouds from all the tunnels in the Seg2Tunnel dataset are used as the training set, the mIoU reaches 90.7 %. The training processes of the five models are plotted in Fig. 12, where it can be found that all models converge within 100 training epochs.

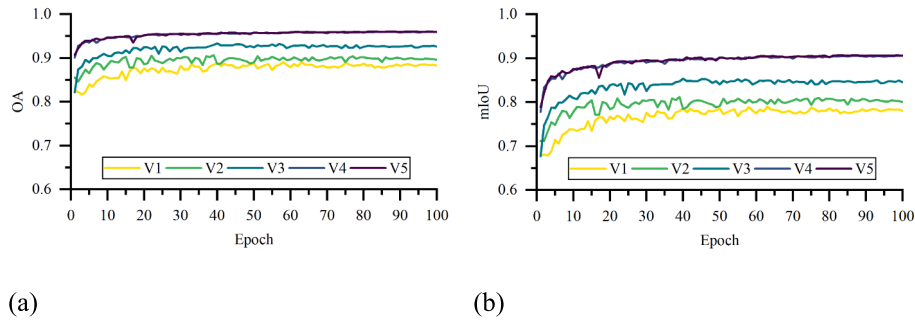


Fig. 12. The training processes of PointNet++ on Level I with different training set sizes according to error metric (a) OA and (b) mIoU on the test set.

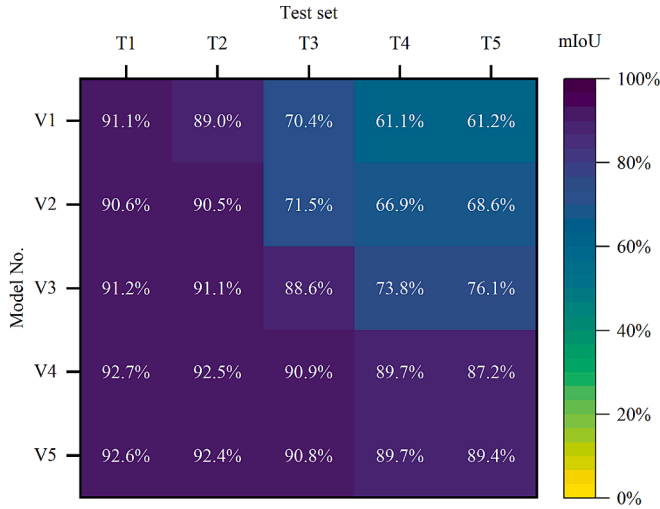


Fig. 13. Detailed mIoU results for PointNet++ applied to Level I data with different training set sizes.

The mIoU of Model V1 to Model V5 are explored in more detail in Fig. 13. For each part of the test set, the mIoU is improved with the increase in the training set volume. Taking tunnel T5 in the test set as an example, although the training sets for Model V1 to Model V4 do not include point clouds from T5, the segmentation performance is still improved with the expansion of the training set, showing promising generalisability. Another important finding is that more diverse training sets do not harm the segmentation performance on any part of the test set. Segmentation results for Model V5 are randomly extracted for visualisation in Fig. 14 and are in line with the present OA and mIoU, where it is shown that most non-structural elements and noise are distinguished from the structural segments well.

5.2. Level II experiment

5.2.1. Data augmentation strategy

Eight experiments are adopted to validate and investigate the effectiveness of the data augmentation strategies. Table 6 and Fig. 15 present the configurations, training process, and metrics of the eight experiments. The training set size is also listed in Table 6 since it is affected by the splitting count factor. As the splitting count factor increases, the model performance shows significant improvement with mIoU reaching 88.6 % for Model A5. Fig. 15 also shows that training converges within fewer training epochs when the splitting count factor is increased. However, the most popular augmentation strategy, 3D rotation (Qi et al., 2017), fails to boost the model performance for Seg2Tunnel. This degradation in performance may be caused by unrealistic point clouds generated by the 3D rotation whereas 2D rotation



Fig. 14. Segmentation results of the PointNet++ Model V5 applied to rings T1-12-637, T3-1-112, and T4-17-607 taken from the test set (arranged top-bottom respectively): (a) ground truth and (b) segmentation result.

generates Cartesian coordinates that are more in line with the real cases.

The confusion matrix of Model A5 is provided in Fig. 16. For a given label, the points correctly predicted vary between 87.54 % and 97.26 %. However, some inaccuracies are also evident, particularly for the segmentation of points with label 4 because these segments are most

Table 6

Model performance of PointNet++ on Level II with different data augmentation strategies.

Model No.	Rotation	Splitting count factor	Training set volume	OA	mIoU
A1	2D	1	591	0.893	0.816
A2	2D	3	1,632	0.922	0.858
A3	2D	6	2,906	0.928	0.864
A4	2D	12	4,937	0.930	0.874
A5	2D	18	6,635	0.940	0.886
A6	3D	6	2,906	0.907	0.824
A7	3D	12	4,937	0.913	0.837
A8	3D	18	6,635	0.924	0.853

The bold font denotes the highest metric in a set of data experiment.

commonly obscured during scanning, as shown in Fig. 6. The confusion matrix shows that the points with label 4 are prone to be falsely predicted as label 0 (background) or label 3 and label 5 (adjacent segments).

5.2.2. Training set size

Similar to Level I, to investigate the effects of the training set diversity, eight models are trained with different training sets. As listed in Table 7, the model performance is improved with increasing size and diversity of the training set. It is demonstrated that sufficient training data can greatly boost model performance before plateauing and eventually reaching a point of diminishing returns.

The segmentation results of Model V1, Model V3, and Model V8 are visualized in Fig. 17. Model V1 provides only obscure predictions of segment boundaries. While Model V3 provides sharp boundary predictions, they are often positioned inaccurately. The segmentation results of Model V8 show not only sharp boundaries but also true positions indicating that accurate and reliable predictions are achievable provided sufficient training data is available.

5.2.3. Input size

Large point clouds provide more features for learning DL network weights during training. To investigate the effects of the input size, seven experiments are considered as listed in Table 8. Comparing Model S1 and Model S7, the mIoU is improved by 5.18 % using the larger input size. Thus, whilst an increase in the input size improves the segmentation results, the improvement is relatively minor compared to the influence of the training set sizes (i.e. number of point clouds).

5.2.4. Hyperparameter

Another seven models are trained with different hyperparameters, as listed in Table 9. In each layer of PointNet++, the features of neighbouring points are aggregated into centres of point groups (Qi et al., 2017). Among the hyperparameters, the radius and number of neighbours determine the receptive field of the point group centres by controlling the aggregation process. Model H1 to Model H4 correspond to four neighbour grouping strategies shown in Fig. 18. Comparing Model

H1 to Model H4, a decrease in the neighbour radius boosts the model performance whereas a decrease in the neighbour number has the opposite effect. Hence, the grouping strategy involving a small radius and a small number of neighbours provides the best performance. This is likely because local features are essential for the segmentation of the tunnel point clouds and the scale of local features is substantially smaller relative to the whole point cloud. The small-radius-small-number strategy presents the most condensed receptive fields. The visualised segmentation results in Fig. 19 indicate that the hyperparameters have a significant influence on the segmentation performance near the segment boundaries. From the performance comparison of Model H3, Model H5, Model H7, and Model H8, the model with the larger MLP size achieves better segmentation results.

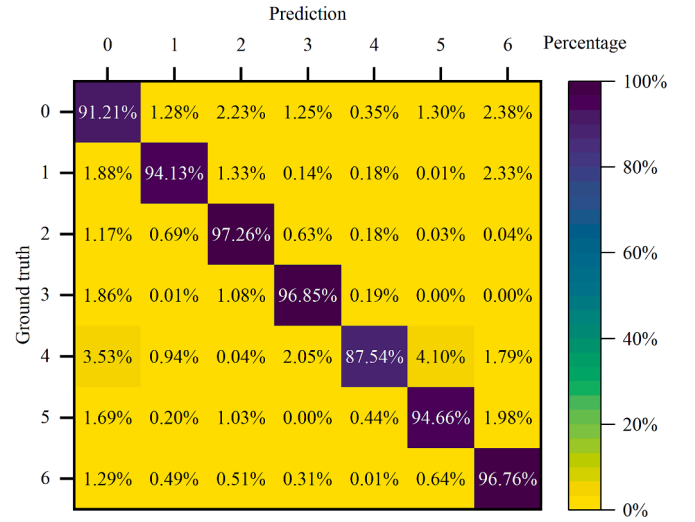


Fig. 16. The confusion matrix of PointNet++ Model A5 for the different labels on Level II.

Table 7

Model performance of PointNet++ on Level II with different training set sizes.

Model No.	Training set	Training set size	OA	mIoU
V1	1/14 T1	177	0.575	0.367
V2	3/14 T1	534	0.829	0.712
V3	7/14 T1	1,234	0.885	0.790
V4	10/14 T1	1,755	0.912	0.836
V5	14/14 T1	2,479	0.924	0.859
V6	14/14 T1 and 5/11 T2	3,126	0.931	0.869
V7	14/14 T1 and 8/11 T2	3,652	0.935	0.876
V8	14/14 T1 and 11/11 T2	4,363	0.942	0.888

The bold font denotes the highest metric in a set of data experiment.

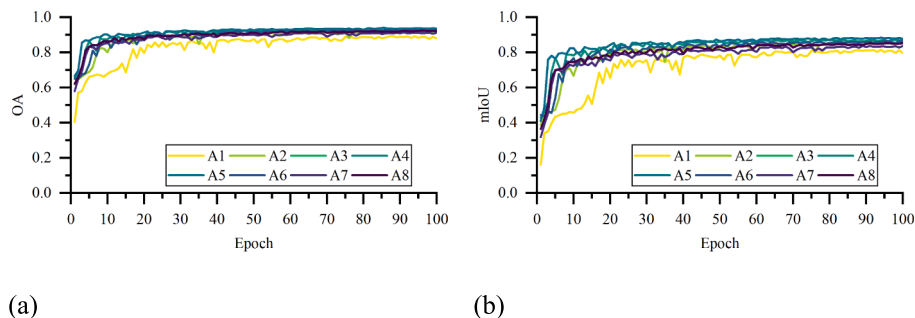
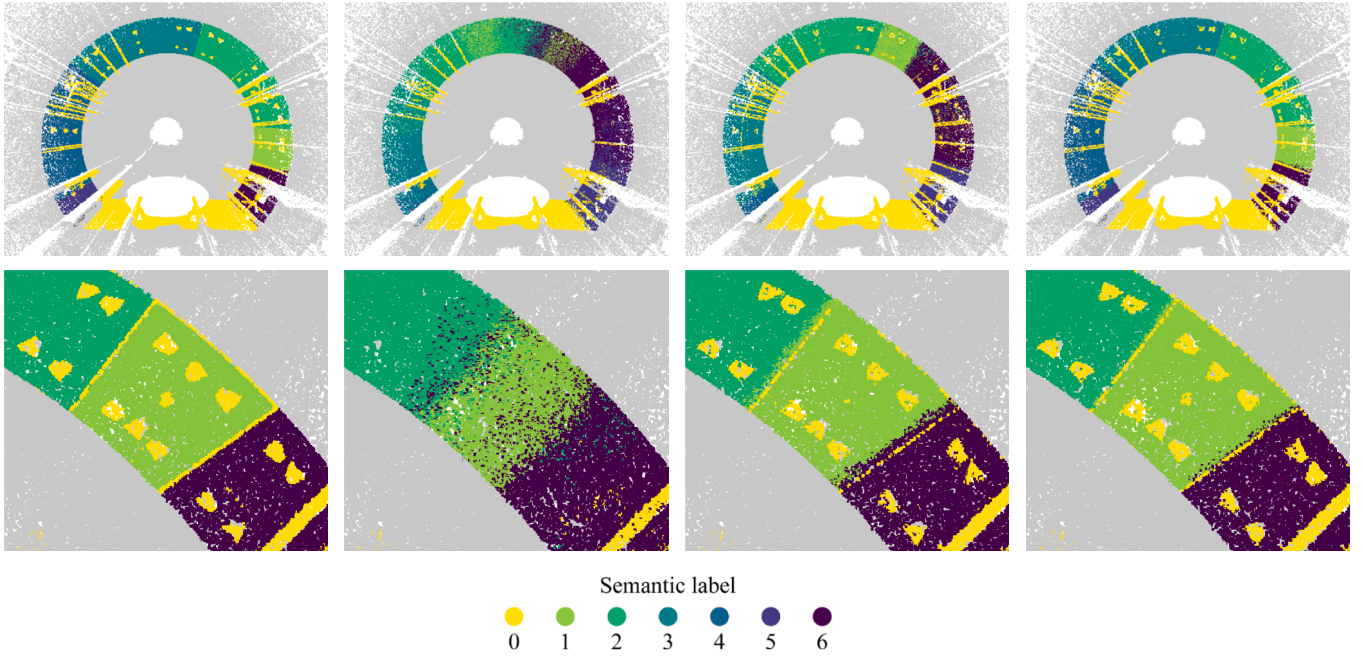


Fig. 15. The training processes of PointNet++ on Level II with different data augmentation strategies according to error metric (a) OA and (b) mIoU on the test set.



(a)

(b)

(c)

(d)

Fig. 17. Segmentation results of PointNet++ on Level II with different training set sizes applied to ring T1-4-199 and ring T1-4-197 taken from the test dataset (arranged top-bottom respectively): (a) ground truth, (b) Model V1, (c) Model V3, and (d) Model V8.

Table 8

Model performance results of PointNet++ on Level II with different input sizes.

Model No.	Input size	OA	mIoU
S1	2,048	0.916	0.850
S2	4,096	0.932	0.873
S3	8,192	0.930	0.874
S4	12,288	0.942	0.888
S5	16,384	0.941	0.884
S6	20,480	0.946	0.894
S7	24,576	0.947	0.894

The bold font denotes the highest metric in a set of data experiment.

5.3. Level III experiment

5.3.1. Input size

Again, five models are trained using different input sizes, as listed in Table 10. The results show that when the input size exceeds 51,200, there is a plateau in model performance. The mIoU of Model S5 reaches 82.8 %, which is acceptable even though the performance deteriorates compared to the Level II results. In Fig. 20, it is shown that ~12 % of points with label 4 are falsely predicted as points with label 3 (adjacent segment), which is similar to the results on Level II.

5.3.2. Hyperparameter

The influence of neighbour numbers is explored using five experiments as described in Table 11. The results show that the neighbour number parameter has a notable influence on model performance. From the visualized segmentation results in Fig. 21, the models trained with relatively small neighbour numbers provide inaccurate predictions near segment boundaries. Predictions are improved using larger neighbour numbers. However, similar to the Level II results, a larger neighbour number does not always provide the best segmentation results.

Table 9

Model performance results of PointNet++ on Level II with different hyperparameters.

Model No.	Layer	Neighbour radius and neighbour number	MLP size	OA	mIoU
H1	1	0.01, 320.02, 640.04, 128	32, 32, 6464, 64, 12864, 96, 128	0.949	0.900
	2	0.04, 640.08, 128	128, 128, 256128, 196, 256		
H2	1	0.01, 640.02, 1280.04, 256	32, 32, 6464, 64, 12864, 96, 128	0.947	0.900
	2	0.04, 1280.08, 256	128, 128, 256128, 196, 256		
H3	1	0.1, 320.2, 640.4, 128	32, 32, 6464, 64, 12864, 96, 128	0.930	0.874
	2	0.4, 640.8, 128	128, 128, 256128, 196, 256		
H4	1	0.1, 640.2, 1280.4, 256	32, 32, 6464, 64, 12864, 96, 128	0.937	0.882
	2	0.4, 1280.8, 256	128, 128, 256128, 196, 256		
H5	1	0.1, 320.2, 640.4, 128	48, 48, 7272, 72, 12872, 96, 128	0.934	0.879
	2	0.4, 640.8, 128	144, 144, 256144, 196, 256		
H6	1	0.1, 320.2, 640.4, 128	48, 48, 8080, 80, 14480, 112, 144	0.936	0.881
	2	0.4, 640.8, 128	196, 196, 272144, 212, 272		
H7	1	0.1, 320.2, 640.4, 128	64, 64, 128128, 128, 256128, 192, 256	0.936	0.882
	2	0.4, 640.8, 128	256, 256, 512256, 392, 512		

The bold font denotes the highest metric in a set of data experiment.

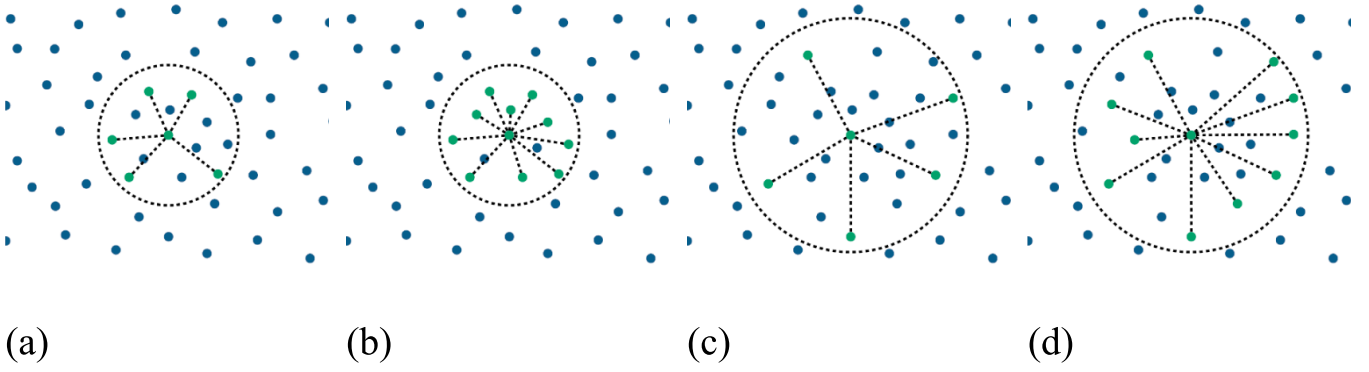


Fig. 18. The neighbour grouping strategies considered in this study: (a) small neighbour radius and small neighbour number (Model H1), (b) small neighbour radius and large neighbour number (Model H2), (c) large neighbour radius and small neighbour number (Model H3), and (d) large neighbour radius and large neighbour number (Model H4).

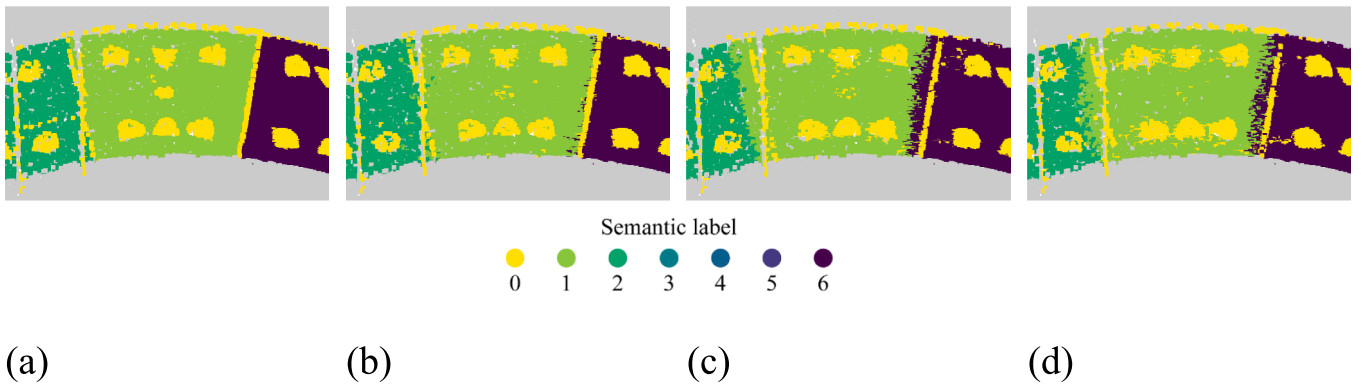


Fig. 19. Segmentation results of PointNet++ on Level II with different hyperparameters applied to ring T1-4-200 taken from the test set: (a) ground truth, (b) Model H1, (c) Model H3, and (d) Model H4.

Table 10

Model performance results of RandLA-Net on Level III with different input sizes.

Model No.	Input size	OA	mIoU
S1	30,720	0.916	0.797
S2	40,960	0.922	0.810
S3	51,200	0.928	0.823
S4	61,400	0.930	0.827
S5	71,680	0.930	0.828

The bold font denotes the highest metric in a set of data experiment.

5.4. Level IV experiment

The segmentation performance of both Mask3D and TD3D is presented in Table 12. For the joint semantic and instance segmentation of the whole point clouds (Task 1), the mAP of TD3D is 27.0 %, showing better results than Mask3D. However, the poor metrics indicate that Task 1 is not readily achievable using these state-of-the-art networks. For Task 2, TD3D shows surprisingly good metrics with mAP reaching 99.0 %, outperforming Mask3D. To further illustrate the segmentation results, TD3D is compared with three unsupervised clustering algorithms for Task 2, as shown in Fig. 22. The colours in Fig. 22 denote different instances. It can be found that the unsupervised clustering algorithms all present poor segmentation results due to the lack of domain applicability learned from the data. In contrast, the TD3D model, trained with Seg2Tunnel, gives satisfying segmentation results.

Given that there is no feasible network for Task 1 and TD3D shows promising performance for Task 2, a two-stage framework based on RandLA-Net and TD3D is proposed. First, the point clouds are input into the trained RandLA-Net model for semantic segmentation. Then, the

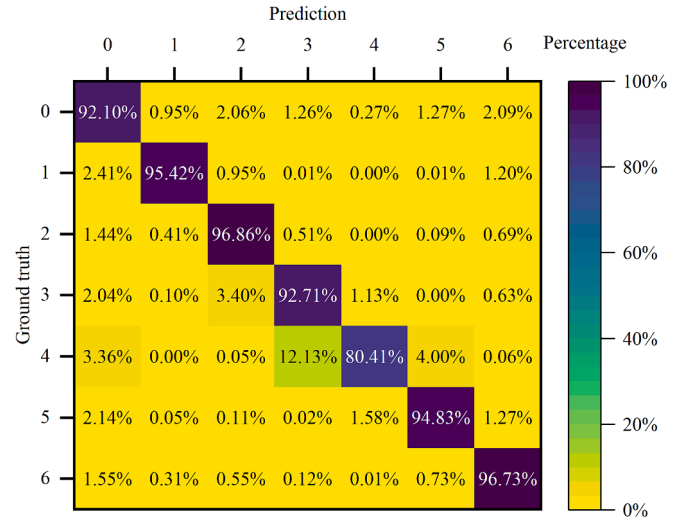


Fig. 20. The confusion matrix of RandLA-Net Model S5 for the different labels on Level III.

point clouds with individual semantic labels are separately supplied to train TD3D models for instance segmentation. Exemplar segmentation results are shown in Fig. 23, where most segments are segmented correctly. Some false predictions occur due to falsely predicted semantic labels. Since the instance segmentation results of the trained TD3D model achieve 99.0 % mAP, the segmentation performance of the proposed two-stage framework is largely dependent on the first stage, i.e.,

Table 11

Model performance results of RandLA-Net on Level III with different hyperparameters.

Model No.	Neighbour number	OA	mIoU
H1	4	0.813	0.584
H2	8	0.899	0.760
H3	16	0.922	0.810
H4	24	0.931	0.831
H5	28	0.932	0.830

The bold font denotes the highest metric in a set of data experiment.

117, Model V1 presents poor segmentation performance (mIoU of 0.396, 0.434, and 0.452, respectively), accounting for the noticeable discrepancies in the deformation calculations. This is likely because Model V1 is not sufficiently trained (only using tunnel T1 point clouds in the training set), which results in unstable segmentation and deformation calculation. The model with better segmentation performance shows more accurate deformation calculation results. For Model V5, the average error and average relative error of the calculated convergence are 0.4 mm and 2.38 % respectively, which is acceptable for structural analysis (Wang et al., 2021). The elliptical fitting result of T5-1-119 is provided

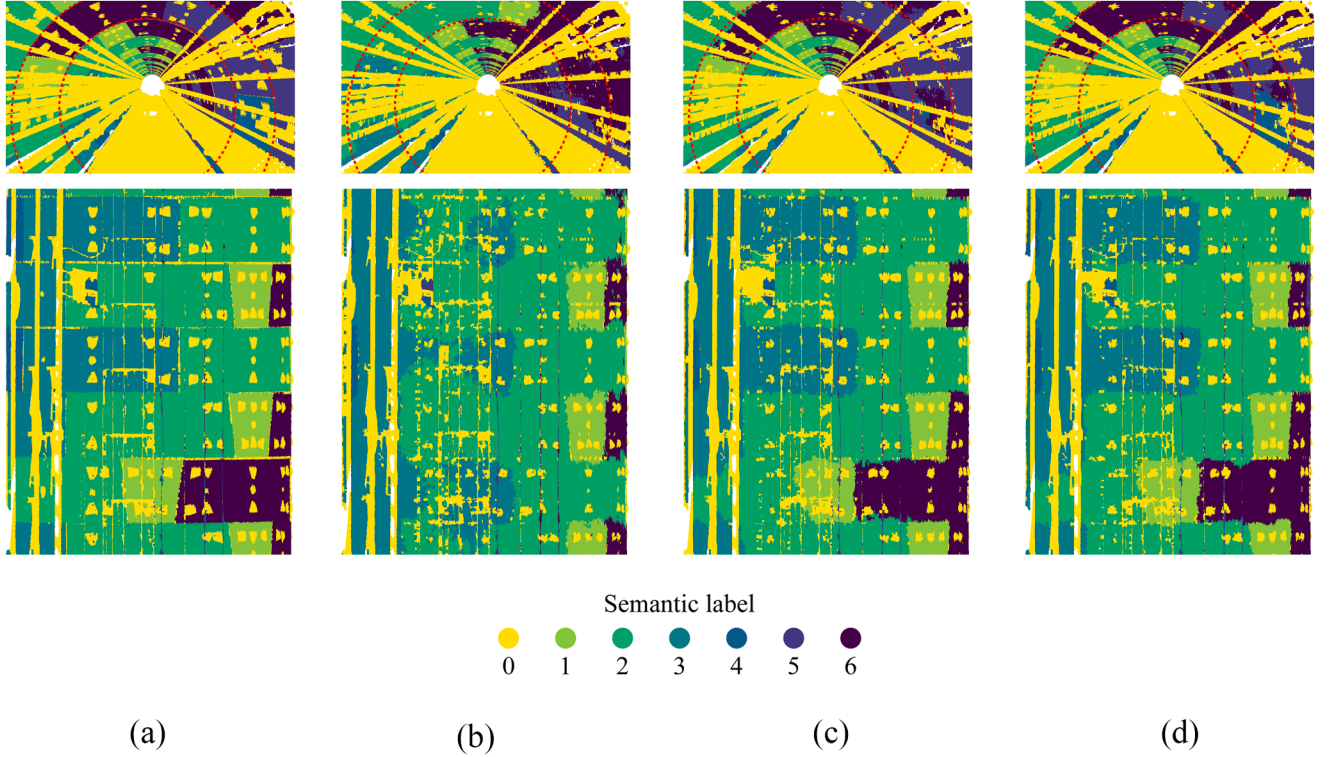


Fig. 21. Segmentation results of RandLA-Net on Level III with different hyperparameters applied to scene T1-15 taken from the test set: (a) ground truth, (b) Model H1, (c) Model H3, and (d) Model H5.

Table 12

Model performance of Mask3D and TD3D on Level IV with different tasks.

Model	Task	mAP	mAP@50	mAP@25
Mask3D	Task 1	0.019	0.069	0.336
TD3D	Task 1	0.270	0.336	0.341
Mask3D	Task 2	0.023	0.079	0.191
TD3D	Task 2	0.990	1.000	1.000

The bold font denotes the highest metric in a set of data experiment.

the semantic segmentation performance of the trained RandLA-Net model.

5.5. Deformation calculation

By way of example, ring-wise deformations are automatically calculated based on the segmentation results of Model V1, Model V3, and Model V5 on Level I. The calculated convergence of segmental lining rings in scene T5-1 is shown in Fig. 24. For all three models, most (80 %) relative errors of the calculated convergence are less than 10.57 %. However, there are some observed discrepancies for Model V1, due to reduced segmentation performance. For T5-1-115, T5-1-116, and T5-1-

in Fig. 25 (the visualised deformation is enlarged 10 times for illustration). During the elliptical fitting, all DL-predicted segment points per ring are used to calculate the deformed shapes based on the least squares method. Hence, while there are a small number of outliers observed in Fig. 25(b) (FP accounts for 1.01 % of TP and FP in this case), the deformation calculation based on segmentation results is robust because of the fitting process.

6. Advantages and limitations

A comparison between the proposed 3D point-based segmentation method (represented by Level II Model H1) and two previously documented benchmarks — an image-based method (Zhang et al., 2022b) and a voxel-based method (Ji et al., 2022) — is shown in Table 13. When comparing the segmentation performance for segment points versus non-segment points, it is observed that the proposed 3D point-based DL method achieves state-of-the-art performance.¹ It is therefore demonstrated that the end-to-end 3D DL models trained by the Seg2Tunnel dataset are superior for application to tunnel point cloud segmentation.

¹ To the authors' knowledge, no other benchmarks exist for higher level segmentation tasks.

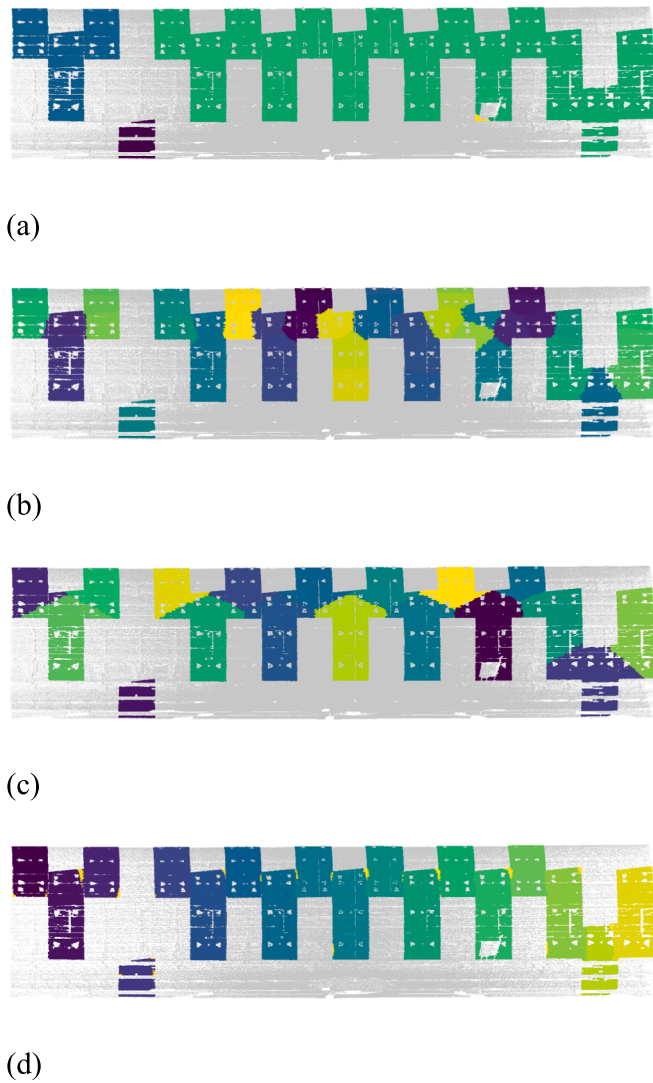


Fig. 22. Segmentation results of unsupervised clustering algorithms and TD3D on Level IV applied to scene T2-14 taken from the test set: (a) DBSCAN, (b) Affinity Propagation, (c) K-means, and (d) TD3D, where different colours denote different instances.

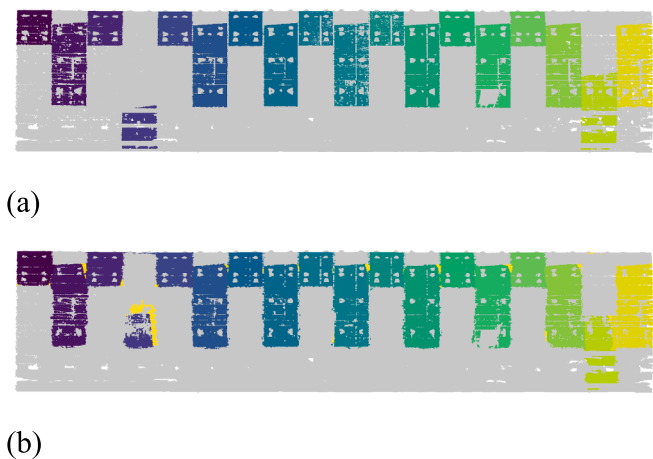


Fig. 23. Segmentation results of the two-stage framework on Level IV applied to scene T2-14 taken from the test set: (a) ground truth and (b) segmentation result, where different colours denote different instances.

The manual segmentation of Level II ring-wise point cloud takes ~ 15 min per ring for experienced annotators. As a comparison, the automated segmentation using the trained 3D DL models (Level II Model V8 on an NVIDIA A100-SXM-80 GB GPU) takes ~ 0.3 s, showcasing significant improvement in efficiency and automation. For Level III scene-wise point cloud (~ 20 rings), the inference time is ~ 1 s and the training takes 6.5 h (Level III Model S5 on an NVIDIA A100-SXM-80 GB GPU). However, increasing the number of points will significantly increase the computational time and memory. For RandLA-Net when handling 102,400 points, the number of learnable parameters and floating-point operations are 4.98 M and 21.35 G respectively, even though RandLA-Net is designed for large-scale point clouds. Hence, future works should consider improvements to the computational efficiency.

Although the proposed dataset and trained models present promising segmentation performance, there are some instances of poor performance for uncommon rings. For example, the uncommon ring displayed in Fig. 26, of which the segment layout is different from the majority in the training dataset, is incorrectly predicted by Level III Model H4 (the best-performing model on Level III). The trained model had not ‘seen’ similar data during training and, as a result, generated some false predictions. The result indicates that there remains room for improvement in both the performance and generalisation of the trained models. This is likely due to three factors: (1) the segments are highly similar and closely connected, (2) the adopted 3D DL networks lack architecture designs tailored to the geometric nature of segmental tunnel linings, and (3) the training dataset lacks sufficient diversity. Hence, future works should focus on tackling these challenges, such as adopting transfer learning approaches on synthetic data or large generic datasets. Another limitation of the proposed dataset is that the defects, such as water leakage and concrete spalling, are not considered and annotated yet. Future efforts can be put into investigating the application of 3D DL to defect segmentation.

7. Conclusions

A unique, diverse, and richly annotated dataset, ‘Seg2Tunnel’, was proposed to boost the application of 3D DL in automatically understanding structural elements, namely segments, in the tunnel point clouds. Seg2Tunnel is composed of segmental lining point clouds acquired from five tunnels, to dilate dataset diversity. A hierarchical dataset architecture was proposed to enable rigorous evaluation of future 3D DL model performance, involving semantic segmentation and instance segmentation. The 3D DL models trained with the Seg2Tunnel dataset showed feasible and promising segmentation performance for each of the four levels considered. The characteristics of class imbalance and shape similarity in the Seg2Tunnel dataset were revealed through quantitative analysis. Through the comprehensive data experiments, the feasibility of applying 3D DL models to segmental lining point clouds was proven and the model performance was widely investigated, providing the benchmarks of the Seg2Tunnel dataset.

Through the data experiments on Level I to Level III, the influence of training set size, data augmentation strategy, input size, and hyperparameter on trained model performance was explored. The training set sizes were shown to have a significant influence on model performance, highlighting the importance of a diverse training set. Given the annotated point clouds, data augmentation using 2D rotation was shown to further improve the segmentation performance of 3D DL models. The segmentation performance of 3D DL models, especially around the segment boundaries, relied on careful selection of the 3D DL network hyperparameters. For both PointNet++ and RandLA-Net, the hyperparameters (neighbour radius and neighbour number) related to the receptive field can greatly affect the segmentation performance. For ring-wise point clouds (using PointNet++), a relatively small receptive field helps to improve the finesse of segmentation performance whilst a relatively large one performs better for scene-wise point clouds (using

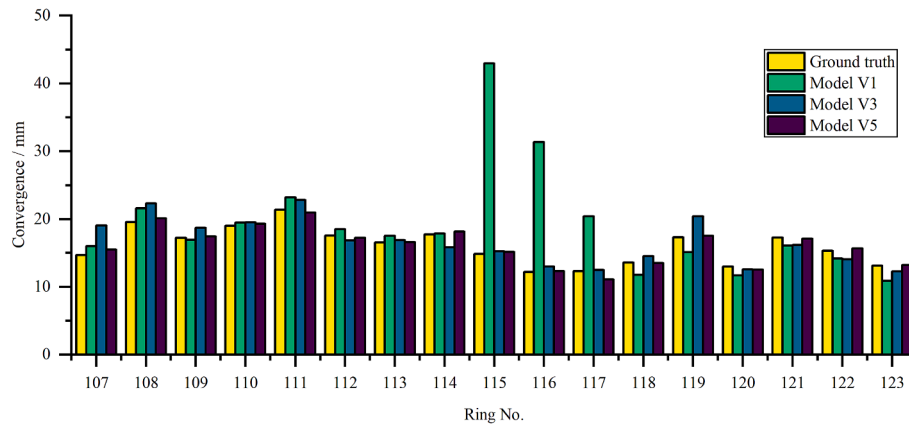


Fig. 24. The calculated convergence of segmental lining rings in scene T5-1.

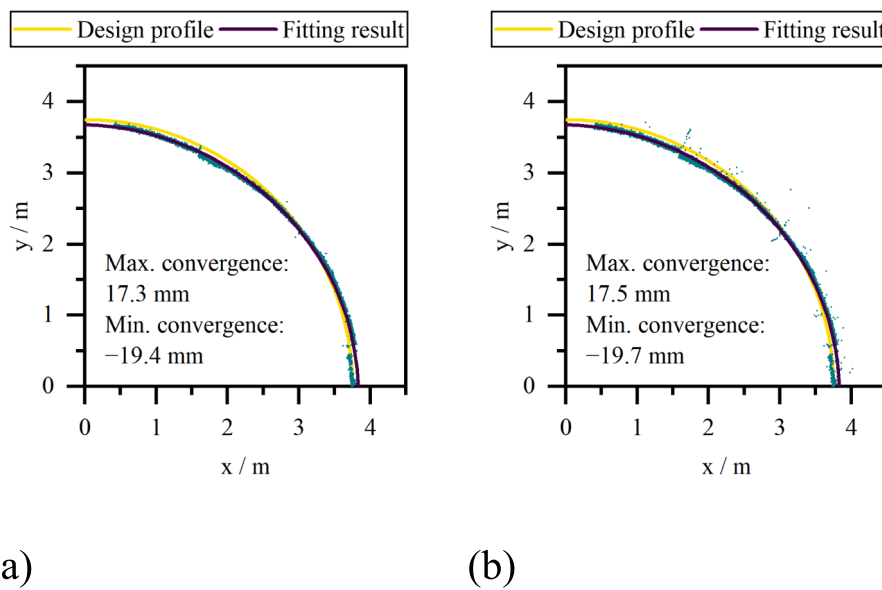


Fig. 25. The elliptical fitting results of the segmental lining ring (T5-1-119): (a) Ground truth and (b) Model V5.

Table 13

The comparison between the proposed 3D point-based segmentation method, image-based method, and voxel-based method.

Method	OA	mIoU
Image-based DL method (Zhang et al., 2022b)	0.852	0.735
Voxel-based DL method (Ji et al., 2022)	0.977	0.926
Proposed 3D point-based DL segmentation method (Level II Model H1)	0.986	0.955

The bold font denotes the highest metric in a set of data experiment.

RandLA-Net). The best mIoU achieved on Level I to Level III is 0.907, 0.900, and 0.831 respectively.

On Level IV, two state-of-the-art 3D DL networks failed to realise the instance segmentation of the whole point cloud. However, the model TD3D provided promising performance on the simpler task of instance segmentation of the partial point clouds with individual labels. A novel two-stage framework based on RandLA-Net (semantic segmentation module) and TD3D (instance segmentation module) was proposed for the fully automatic instance segmentation of large-scale segmental lining point clouds, demonstrating satisfactory segmentation results. It is considered to develop an end-to-end architecture based on the proposed framework in future studies.

The application of 3D DL to segment segmentation provides a fully automatic workflow from scanning to analysis. The automatically calculated deformation was demonstrated to be accurate with an average relative error of 2.38 %. This study established the feasibility of applying 3D DL to segment segmentation for the first time, supported by the Seg2Tunnel dataset. It is indicated that the Seg2Tunnel dataset is a firm data foundation to foster the development of new customised 3D DL networks, ultimately paving the way for intelligent analysis of segmental tunnel lining point clouds and efficient inspection of large-scale tunnel assets.

CRediT authorship contribution statement

Wei Lin: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Software, Validation, Visualization, Writing – original draft. **Brian Sheil:** Conceptualization, Formal analysis, Methodology, Project administration, Resources, Supervision, Writing – review & editing. **Pin Zhang:** Conceptualization, Methodology, Supervision, Writing – review & editing. **Biao Zhou:** Data curation, Validation, Visualization, Writing – review & editing. **Cheng Wang:** Data curation, Project administration, Validation. **Xiongyao Xie:** Conceptualization, Funding acquisition, Investigation, Project administration, Resources, Validation, Writing – review & editing.

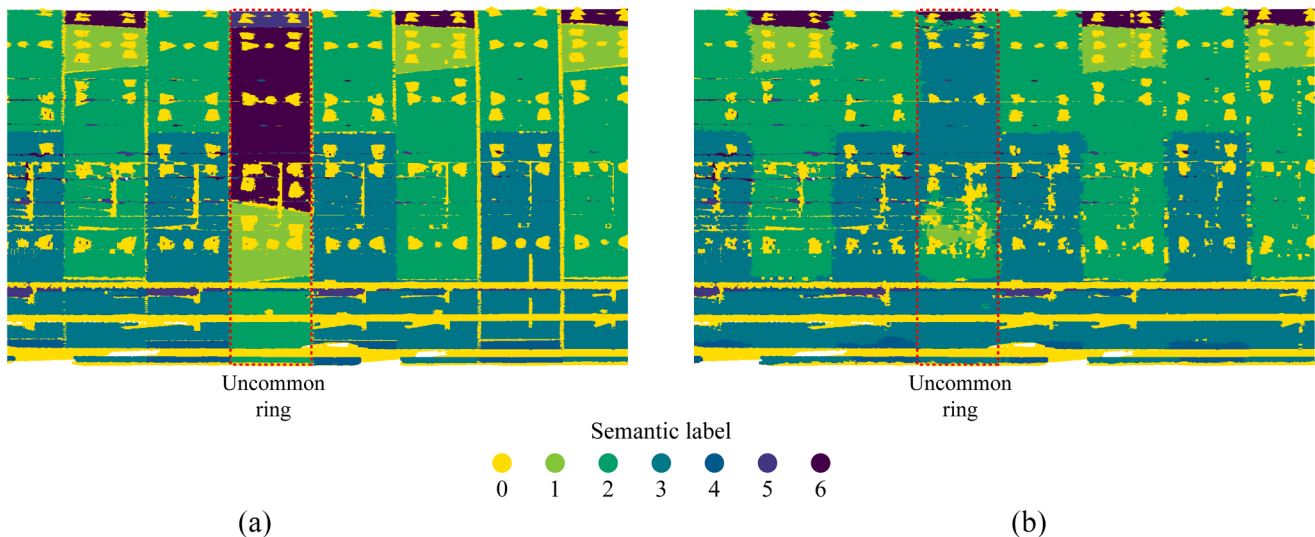


Fig. 26. An example of a poor segmentation performance obtained using Level III Model H4: (a) ground truth and (b) segmentation result.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgements

The authors appreciate Zhen Chen at Peking University, Yixiong Jing at Oxford University, and Yiting Li at Tongji University for their contributions to this work.

This work was supported by the National Key R&D Program of China (Grant Numbers 2023YFC3806701, 2023YFC3806702, 2023YFC3806705), the National Natural Science Foundation of China [grant numbers 52038008, 52378408], the China Scholarship Council [grant number 202206260174], the Science and Technology Commission of Shanghai Municipality [grant number 22DZ1203004], and the State Grid Shanghai Municipal Electric Power Company [grant number 52090W23000B].

This work was performed using resources provided by the Cambridge Service for Data Driven Discovery (CSD3) operated by the University of Cambridge Research Computing Service (www.csd3.cam.ac.uk), provided by Dell EMC and Intel using Tier-2 funding from the Engineering and Physical Sciences Research Council (capital grant EP/T022159/1), and DIRAC funding from the Science and Technology Facilities Council (www.dirac.ac.uk).

References

- Armeni, I., Sax, S., Zamir, A.R., Savarese, S., 2017. Joint 2D-3D-semantic data for indoor scene understanding. doi: 10.48550/arXiv.1702.01105.
- Behley, J., Garbade, M., Milioto, A., Quenzel, J., Behnke, S., Gall, J., Stachniss, C., 2021. Towards 3D LiDAR-based semantic scene understanding of 3D point cloud sequences: the SemanticKITTI Dataset. *Int. J. Rob. Res.* 40, 959–967. <https://doi.org/10.1177/02783649211006735>.
- Chang, A.X., Funkhouser, T., Guibas, L., Hanrahan, P., Huang, Q., Li, Z., Savarese, S., Savva, M., Song, S., Su, H., Xiao, J., Yi, L., Yu, F., 2015. ShapeNet: an information-rich 3D model repository. doi: 10.48550/arXiv.1512.03012.
- Chang, J., Huang, H., Zhang, D., Wu, H., Yan, J., 2022. Transverse deformational behaviors of segmental lining during shield tunneling: a case study. *Struct. Control Heal. Monit.* 29, e3097.
- Chang, J., Zhang, D., Huang, H., Jia, J., 2023. A hybrid sensing of rotation-induced stress of segmental lining during shield tunneling via WSN and surrogate numerical modeling. *Tunn. Undergr. Sp. Technol.* 140, 105315 <https://doi.org/10.1016/j.tust.2023.105315>.
- Chen, R.-P., Chen, S., Wu, H.-N., Liu, Y., Meng, F.-Y., 2020. Investigation on deformation behavior and failure mechanism of a segmental ring in shield tunnels based on elaborate numerical simulation. *Eng. Fail. Anal.* 117, 104960 <https://doi.org/10.1016/j.engfailanal.2020.104960>.
- Chen, C., Li, G., Xu, R., Chen, T., Wang, M., Lin, L., 2019. Clusternet: deep hierarchical cluster network with rigorously rotation-invariant representation for point cloud analysis. In: 2019 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), pp. 4989–4997. <https://doi.org/10.1109/CVPR.2019.00513>.
- Cheng, R., Zeng, H., Zhang, B., Wang, X., Zhao, T., 2023. FFA-Net: fast feature aggregation network for 3D point cloud segmentation. *Mach. Vis. Appl.* 34, 80. <https://doi.org/10.1007/s00138-023-01434-x>.
- Cui, H., Ren, X., Mao, Q., Hu, Q., Wang, W., 2019. Shield subway tunnel deformation detection based on mobile laser scanning. *Autom. Constr.* 106, 102889 <https://doi.org/10.1016/j.autcon.2019.102889>.
- Deng, S., Dong, Q., 2021. GA-NET: global attention network for point cloud semantic segmentation. *IEEE Signal Process. Lett.* 28, 1300–1304. <https://doi.org/10.1109/LSP.2021.3082851>.
- Du, L., Zhong, R., Sun, H., Pang, Y., Mo, Y., 2022. Dislocation detection of shield tunnel based on dense cross-sectional point clouds. *IEEE Trans. Intell. Transp. Syst.* 23, 22227–22243. <https://doi.org/10.1109/TITS.2022.3156385>.
- Duan, D.-Y., Qiu, W.-G., Cheng, Y.-J., Zheng, Y.-C., Lu, F., 2021. Reconstruction of shield tunnel lining using point cloud. *Autom. Constr.* 130, 103860 <https://doi.org/10.1016/j.autcon.2021.103860>.
- Everingham, M., Van Gool, L., Williams, C.K.I., Winn, J., Zisserman, A., 2010. The pascal visual object classes (VOC) challenge. *Int. J. Comput. Vis.* 88, 303–338. <https://doi.org/10.1007/s11263-009-0275-4>.
- Goyal, M., Murugappan, S., Piya, C., Benjamin, W., Fang, Y., Liu, M., Ramani, K., 2012. Towards locally and globally shape-aware reverse 3D modeling. *Comput. Des.* 44, 537–553. <https://doi.org/10.1016/j.cad.2011.12.004>.
- Hackel, T., Wegner, J.D., Savinov, N., Ladicky, L., Schindler, K., Pollefeys, M., 2018. Large-scale supervised learning for 3D point cloud labelling: Semantic3d.Net. *Photogramm. Eng. Remote Sensing* 84, 297–308. <https://doi.org/10.14358/PERS.84.5.297>.
- Hegemann, F., Stascheit, J., Maidl, U., 2020. As-built documentation of segmental lining rings in the BIM representation of tunnels. *Tunn. Undergr. Sp. Technol.* 106, 103582 <https://doi.org/10.1016/j.tust.2020.103582>.
- Hu, Q., Yang, B., Xie, L., Rosa, S., Guo, Y., Wang, Z., Trigoni, N., Markham, A., 2022. Learning semantic segmentation of large-scale point clouds with random sampling. *IEEE Trans. Pattern Anal. Mach. Intell.* 44, 8338–8354. <https://doi.org/10.1109/TPAMI.2021.3083288>.
- Huang, H., Li, Q., Zhang, D., 2018. Deep learning based image recognition for crack and leakage defects of metro shield tunnel. *Tunn. Undergr. Sp. Technol.* 77, 166–176. <https://doi.org/10.1016/j.tust.2018.04.002>.
- Huang, M.Q., Ninić, J., Zhang, Q.B., 2021. BIM, machine learning and computer vision techniques in underground construction: current status and future perspectives. *Tunn. Undergr. Sp. Technol.* 103677 <https://doi.org/10.1016/j.tust.2020.103677>.
- Itakura, K., Miyatani, S., Hosoi, F., 2022. Estimating tree structural parameters via automatic tree segmentation from LiDAR point cloud data. *IEEE J. Sel. Top. Appl. Earth Obs. Remote Sens.* 15, 555–564. <https://doi.org/10.1109/JSTARS.2021.3135491>.
- Ji, A., Chew, A.W.Z., Xue, X., Zhang, L., 2022. An encoder-decoder deep learning method for multi-class object segmentation from 3D tunnel point clouds. *Autom. Constr.* 137, 104187 <https://doi.org/10.1016/j.autcon.2022.104187>.

- Ji, A., Zhou, Y., Zhang, L., Tiong, R.L.K., Xue, X., 2023. Semi-supervised learning-based point cloud network for segmentation of 3D tunnel scenes. *Autom. Constr.* 146, 104668 <https://doi.org/10.1016/j.autcon.2022.104668>.
- Jing, Y., Sheil, B., Acikgoz, S., 2022. Segmentation of large-scale masonry arch bridge point clouds with a synthetic simulator and the BridgeNet neural network. *Autom. Constr.* 142, 104459 <https://doi.org/10.1016/j.autcon.2022.104459>.
- Kolodiaznyh, M., Rukhovich, D., Vorontsova, A., Konushin, A., 2023. Top-down beats bottom-up in 3D instance segmentation. doi: 10.48550/arXiv.2302.02871.
- Li, Y., Shi, J., Xiao, Z., 2022. A multilevel point cloud classification method for underground tunnels based on three-dimensional moving LiDAR measurements. *Mob. Inf. Syst.*, 6181182 <https://doi.org/10.1155/2022/6181182>.
- Liebers, C., Prochazka, M., Pfützenreuter, N., Liebers, J., Auda, J., Gruenefeld, U., Schneegass, S., 2023. Pointing it out! Comparing manual segmentation of 3D point clouds between desktop, tablet, and virtual reality. *Int. J. Hum. Comput. Interact.* <https://doi.org/10.1080/10447318.2023.2238945>.
- Lin, T.-Y., Maire, M., Belongie, S., Hays, J., Perona, P., Ramanan, D., Dollár, P., Zitnick, C.L., 2014. Microsoft COCO: common objects in context. In: *Computer Vision – ECCV 2014*. pp. 740–755. doi: 10.1007/978-3-319-10602-1_48.
- Lin, W., Xie, X., Zhou, B., Li, P., Wang, C., 2023b. Refined perception and management of ring-wise deformation information for shield tunnels based on point cloud deep learning and BIM. In: *Eighth International Symposium on Life-Cycle Civil Engineering (IALCCE 2023)*. pp. 3991–3998. doi: 10.1201/9781003323020-490.
- Lin, W., Li, P., Xie, X., 2022a. A novel detection and assessment method for operational defects of pipe jacking tunnel based on 3D longitudinal deformation curve: a case study. *Sensors* 22, 7648. <https://doi.org/10.3390/s22197648>.
- Lin, W., Xie, X., Li, P., Xiao, B., Lu, X., Feng, B., Jin, P., Hu, Y., 2022b. Prediction of settlement induced by tidal fluctuation for underwater shield tunnel during service based on historical monitoring data. In: *2022 8th International Conference on Hydraulic and Civil Engineering: Deep Space Intelligent Development and Utilization Forum (ICHCE)*, pp. 1042–1047. <https://doi.org/10.1109/ICHCE57331.2022.10042697>.
- Lin, W., Li, P., Xie, X., Cao, Y., Zhang, Y., 2023. A novel back-analysis approach for the external loads on shield tunnel lining in service based on monitored deformation. *Struct. Control Heal. Monit.* 2023, 8128701 <https://doi.org/10.1155/2023/8128701>.
- Liu, X., Dong, Z., Bai, Y., Zhu, Y., 2017. Investigation of the structural effect induced by stagger joints in segmental tunnel linings: first results from full-scale ring tests. *Tunn. Undergr. Sp. Technol.* 66, 1–18. <https://doi.org/10.1016/j.tust.2017.03.008>.
- Mahmoudi, M., Sapiro, G., 2009. Three-dimensional point cloud recognition via distributions of geometric distances. *Graph. Model.* 71, 22–31. <https://doi.org/10.1016/j.gmod.2008.10.002>.
- Mo, K., Zhu, S., Chang, A.X., Yi, L., Tripathi, S., Guibas, L.J., Su, H., 2019. PartNet: a large-scale benchmark for fine-grained and hierarchical part-level 3D object understanding. In: *2019 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. pp. 909–918. doi: 10.1109/CVPR.2019.00100.
- Mu, B., Xie, X., Li, X., Li, J., Shao, C., Zhao, J., 2021. Monitoring, modelling and prediction of segmental lining deformation and ground settlement of an EPB tunnel in different soils. *Tunn. Undergr. Sp. Technol.* 113, 103870 <https://doi.org/10.1016/j.tust.2021.103870>.
- Ninic, J., Alsahly, A., Vonthron, A., Bui, H.G., Koch, C., König, M., Meschke, G., 2021. From digital models to numerical analysis for mechanised tunnelling: a fully automated design-through-analysis workflow. *Tunn. Undergr. Sp. Technol.*, 103622.
- Osada, R., Funkhouser, T., Chazelle, B., Dobkin, D., 2001. Matching 3D models with shape distributions. In: *Proceedings International Conference on Shape Modeling and Applications*, pp. 154–166. <https://doi.org/10.1109/SMA.2001.923386>.
- PyTorch documentation [WWW Document], 2023. URL <https://pytorch.org/docs/2.2/> (accessed 2.5.24).
- Qi, C.R., Yi, L., Su, H., Guibas, L.J., 2017. PointNet++: deep hierarchical feature learning on point sets in a metric space. In: *Proceedings of the 31st International Conference on Neural Information Processing Systems*. pp. 5105–5114. doi: 10.5555/3295222.3295263.
- Qian, G., Li, Y., Peng, H., Mai, J., Hammoud, H.A.A.K., Elhoseiny, M., Ghanem, B., 2022. PointNetXt: revisiting PointNet++ with improved training and scaling strategies. In: *Advances in Neural Information Processing Systems 35 (NeurIPS 2022)*.
- Roca-Pardiñas, J., Argüelles-Fraga, R., de Asís López, F., Ordóñez, C., 2014. Analysis of the influence of range and angle of incidence of terrestrial laser scanning measurements on tunnel inspection. *Tunn. Undergr. Sp. Technol.* 43, 133–139. <https://doi.org/10.1016/j.tust.2014.04.011>.
- Schroff, F., Kalenichenko, D., Philbin, J., 2015. FaceNet: a unified embedding for face recognition and clustering. In: *2015 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*. pp. 815–823. doi: 10.1109/CVPR.2015.7298682.
- Schult, J., Engelmann, F., Hermans, A., Litany, O., Tang, S., Leibe, B., 2023. Mask3D: mask transformer for 3D semantic instance segmentation. doi: 10.48550/arXiv.2210.03105.
- Soilán, M., Riveiro, B., Balado, J., Arias, P., 2020. Comparison of heuristic and deep learning-based methods for ground classification from aerial point clouds. *Int. J. Digit. Earth* 13, 1115–1134. <https://doi.org/10.1080/17538947.2019.1663948>.
- Wang, F., Shi, J., Huang, H., Zhang, D., Liu, D., 2021. A horizontal convergence monitoring method based on wireless tilt sensors for shield tunnels with straight joints. *Struct. Infrastruct. Eng.* 17, 1194–1209. <https://doi.org/10.1080/15732479.2020.1801767>.
- Xie, X., Cai, J., Wang, H., Wang, Q., Xu, J., Zhou, Y., Zhou, B., 2022. Sparse-sensing and superpixel-based segmentation model for concrete cracks. *Comput. Civ. Infrastruct. Eng.* 37, 1769–1784. <https://doi.org/10.1111/mice.12903>.
- Xie, J., Huang, X., Zhang, Z., Jin, G., 2023. Cohesive zone model-based analyses of localized leakage of segmentally lined tunnels. *Front. Struct. Civ. Eng.* 17, 503–521. <https://doi.org/10.1007/s11709-023-0927-4>.
- Xie, X., Lu, X., 2017. Development of a 3D modeling algorithm for tunnel deformation monitoring based on terrestrial laser scanning. *Undergr. Sp. Technol.* 2, 16–29. <https://doi.org/10.1016/j.undsp.2017.02.001>.
- Xie, Y., Tian, J., Zhu, X.X., 2020. Linking points with labels in 3D: a review of point cloud semantic segmentation. *IEEE Geosci. Remote Sens. Mag.* 8, 38–59. <https://doi.org/10.1109/MGRS.2019.2937630>.
- Xu, J., Ding, L., Luo, H., Chen, E.J., Wei, L., 2019. Near real-time circular tunnel shield segment assembly quality inspection using point cloud data: a case study. *Tunn. Undergr. Sp. Technol.* 91, 102998 <https://doi.org/10.1016/j.tust.2019.102998>.
- Yan, X., Zhang, H., Wu, Y., Lin, C., Liu, S., 2023. Construction instance segmentation (CIS) dataset for deep learning-based computer vision. *Autom. Constr.* 156, 105083 <https://doi.org/10.1016/j.autcon.2023.105083>.
- Yi, C., Lu, D., Xie, Q., Liu, S., Li, H., Wei, M., Wang, J., 2019. Hierarchical tunnel modeling from 3D raw LiDAR point cloud. *Comput. Des.* 143–154 <https://doi.org/10.1016/j.cad.2019.05.033>.
- Yu, A., Mei, W., Han, M., 2021. Deep learning based method of longitudinal dislocation detection for metro shield tunnel segment. *Tunn. Undergr. Sp. Technol.* 113, 103949 <https://doi.org/10.1016/j.tust.2021.103949>.
- Yu, T., Zhang, Y., Yan, Z., 2023. A new generation method of tunnel progressive defect status random field (TPDSRF) for subway tunnel structure. *Tunn. Undergr. Sp. Technol.* 141, 105340 <https://doi.org/10.1016/j.tust.2023.105340>.
- Zhan, L., Li, W., Min, W., 2023. FA-ResNet: feature affine residual network for large-scale point cloud segmentation. *Int. J. Appl. Earth Obs. Geoinf.* 118, 103259 <https://doi.org/10.1016/j.jag.2023.103259>.
- Zhang, H., Duan, Z., Zheng, N., Li, Y., Zeng, Y., Shi, W., 2022a. An efficient class-constrained DBSCAN approach for large-scale point cloud clustering. *IEEE J. Sel. Top. Appl. Earth Obs. Remote Sens.* 15, 7323–7332.
- Zhang, Z., Ji, A., Wang, K., Zhang, L., 2022b. UnrollingNet: an attention-based deep learning approach for the segmentation of large-scale point clouds of tunnels. *Autom. Constr.* 142, 104456 <https://doi.org/10.1016/j.autcon.2022.104456>.
- Zhou, M., Cheng, W., Huang, H., Chen, J., 2021. A novel approach to automated 3D spalling defects inspection in railway tunnel linings using laser intensity and depth information. *Sensors* 21, 5725.
- Zhou, Y., Luo, H., Yang, Y., 2017. Implementation of augmented reality for segment displacement inspection during tunneling construction. *Autom. Constr.* 82, 112–121. <https://doi.org/10.1016/j.autcon.2017.02.007>.
- Zhou, Y., Wang, Y., Ding, L., Love, P.E.D., 2018. Utilizing IFC for shield segment assembly in underground tunneling. *Autom. Constr.* 93, 178–191. <https://doi.org/10.1016/j.autcon.2018.05.016>.
- Zhou, Y., Ji, A., Zhang, L., Xue, X., 2023. Attention-enhanced sampling point cloud network (ASPCNet) for efficient 3D tunnel semantic segmentation. *Autom. Constr.* 146, 104667 <https://doi.org/10.1016/j.autcon.2022.104667>.
- Zhou, B., Xie, X., Li, Y., 2014. A structural health assessment method for shield tunnels based on torsional wave speed. *Sci. China Technol. Sci.* 57, 1109–1120. <https://doi.org/10.1007/s11431-014-5554-9>.